TOMOSS.A.D.

Master blueprint architecture

Version 0.3 | 30 October 2026 | Architecture, build contracts and reference evidence

TOMOSS.A.D. is defined here as a versioned operating architecture whose active component graph is assembled from a validated state descriptor table, LUT(sS). Its exact numerical foundation is log net D. Jobs execute against coherent snapshots, preserve the meaning of their inputs, and commit traceable results before those results reach displays, networks or devices.

This blueprint consolidates the two supplied PDFs, the earlier log net D specification and the conversation attachment. Version 0.3 adds concrete build interfaces, fixture records and a reproducible C++ reference evaluator to the ECS, compiler and deployment architecture. TOMOSS.A.D. remains the exact project name; no acronym expansion is invented.

Architecture element	Evidence status
log net D numerical foundation	Existing reference package. Its supplied tests and independent checks were reproduced in this conversation.
LUT(sS), ECS and concurrent execution	Proposed normative contracts, grounded in the runtime-diagram, multithreading and modular ECS requirements.
Serial reference evaluator	Complete source and executed checks for the small exact fixture. The scope and measured evidence appear in Sections 66-72.
Physics, rendering, AI and edge integration	Optional modules with explicit contracts. No integrated implementation or calibrated physical result is demonstrated.
Deployment boundary	Initial implementation is a user-space runtime on an existing host OS. A bootable kernel and driver stack are outside the supplied implementation.

The central design decision

Compile a descriptor into a complete immutable runtime plan. Bind each admitted job to one plan, one input manifest and one numerical policy. A state transition may select another plan for that instance at a declared boundary. Inspect the graph that was actually instantiated.

The resulting architecture preserves exact data contracts while making dynamic behavior, concurrency and physical interpretation explicit. It is a master implementation blueprint, with the distinction between verified behavior and future work maintained throughout. [M; E; T L755, L892; S; V]

2 Reading guide and authority

How to interpret requirements and evidence

Sections	Purpose
3-9	System map, records, exact numerical core and operator interfaces.
10-19	Runtime assembly, state, concurrency, evidence, AI and mobile foundations.
20-25	Corrected mechanics, thermal and rendering workload; exact transition.
26-37	Recovery, budgets, verified baseline, acceptance, source reconciliation and references.
38-43	Selected defaults; ECS identities, worlds, systems and descriptor v2.
44-51	Native ABI, compiler tiers, model/context limits and durable coordination.
52-57	Telemetry, terrain, rendering, Android gates and the buildable ECS fixture.
58-60	Primary references for the version 0.2 implementation extension.
61-72	Build interfaces, concrete fixture records, complete C++ source and executed reference evidence.
73-77	Evolution contracts, app workflow, resource admission and build acceptance.

Requirement language

MUST and shall identify conformance requirements; SHOULD gives a default requiring a reason for an exception; MAY marks an option. These specify proposed target behaviour, not completed implementation.

Source precedence

The request supplies the name and master-PDF objective. Source correction T L586 retains mechanics while excluding personal identifiers and religious/philosophical semantics. The numerical master governs retained payloads. Generated explanations are design input, not independent validation.

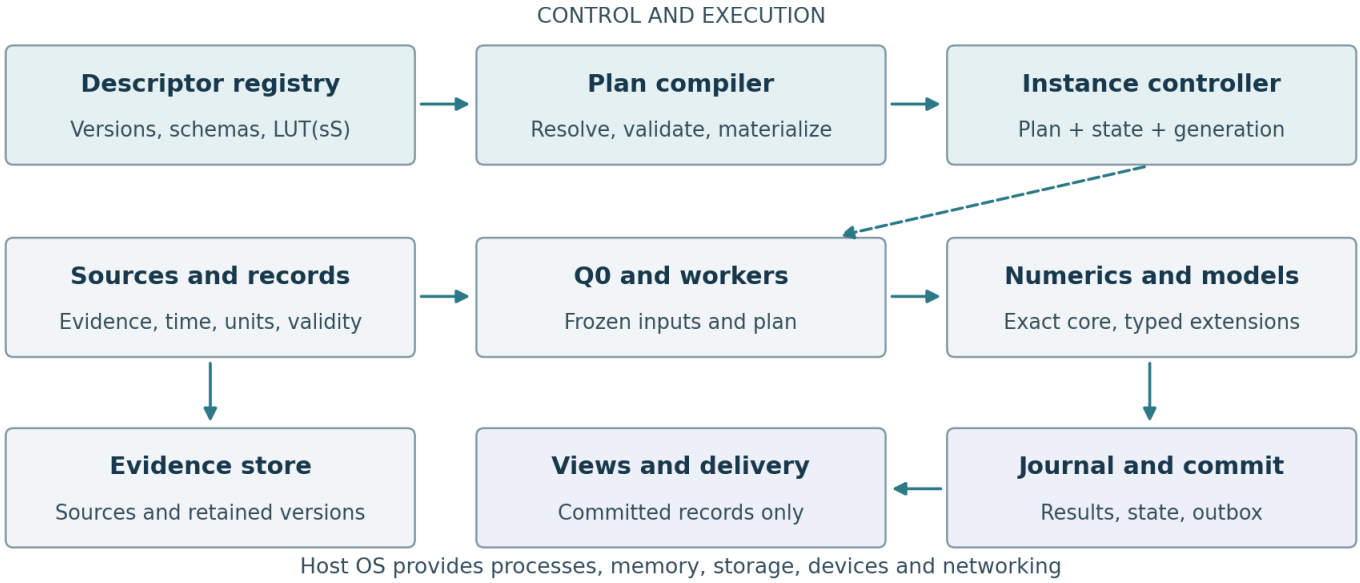
Evidence labels

RETAINED identifies an existing numerical contract. VERIFIED identifies a check actually reproduced in this conversation. PROPOSED identifies a new engineering decision. ILLUSTRATIVE identifies a fixture or worked example. UNRESOLVED identifies a term or decision that the supplied evidence does not settle.

Source keys: [M] numerical master; [E] explanation PDF; [T] attachment text; [S] earlier specification; [V] reproduced core checks. Sections 32-33 reconcile these. [X1-X16] and [Y1-Y13] identify references in Sections 36-37 and 58-60. [V3] is the new reference evidence; [Z1-Z2] appear in Section 77. PDF pages count from the first page.

3 System architecture

Control selects the graph and data moves through it



Proposed TOMOSS.A.D. architecture. The control plane creates versioned plans; the execution path preserves typed data and commits results. Shared services enforce contracts across both.

Module	Owned responsibility
Source and evidence service	Capture source bytes, observations, timing, calibration references and evidence class.
Descriptor registry and plan compiler	Resolve LUT(sS), component versions, schemas, capabilities and dependencies into an immutable plan.
Instance controller	Own each model instance's state version, transition decisions and active published snapshot.
Q0 and workers	Admit bounded jobs, freeze dependencies, execute kernels and prepare results.
log net D and model services	Provide exact numerical kernels and separately profiled domain computations.
Journal and transactional outbox	Persist results, state transitions, provenance and required delivery events together.
Output and host adapters	Render committed views or use explicitly available storage, device and network capabilities.

A document parser, an angle transformation and a beam solver are independent services. A job includes only the components its contracts require. Shared infrastructure does not force every source through every physical or AI model. The host OS owns processes, virtual memory, drivers and network access. [M pp. 12, 19-26; T L713-754; S §§2-7]

4 Governing rules and vocabulary

Stable meaning across dynamic configurations

Rule	Normative contract
Typed meaning	Every value has a schema and numerical profile. Physical values carry units, frame, time and quality as applicable.
Validity	Zero, missing, invalid, unknown and uncalibrated remain distinguishable. Unknown state codes never select a normal-state fallback.
Version identity	Inputs, operators, descriptors, model parameters and backends are identified by immutable version and digest.
Exactness	Declare integer, rational, symbolic, certified interval or approximate computation. Storage precision does not establish physical accuracy.
Bounded behavior	Validate ranges, graph size, queue capacity, memory, precision and permitted side effects before execution.
Coherent execution	An admitted job retains its entire plan and input snapshot. Reconfiguration cannot replace part of an active job's meaning.
Recoverable results	A result, terminal job disposition and required outbox events commit together. Observers see only committed records.
Authority	Reading, computing, rendering, transmitting and commanding a device are separate capabilities. A LUT row does not grant new authority.

Terms with precise meanings

An automorphism is a bijection that preserves a declared structure. D6 acts on the hexagonal structure. A feedback update can be noninvertible and is called a state transition. Polymorphism means registered implementations satisfy a stable interface. Runtime composition binds such implementations into a graph; it need not rewrite language inheritance or virtual tables.

Determinism means one result follows from fixed inputs, state, versions, numerical rules and event order. Bit replay additionally requires a backend with demonstrated bitwise repeatability. Model fidelity and measurement uncertainty are separate questions.

Source terms kept without invented expansions

TOMOSS.A.D., OOB and AC remain unexpanded. In this blueprint, sS is a proposed name for an instance's discrete model state; it is not a system-wide switch. BIG 5 refers to the five patterns selected in Section 15. Q0 names the bounded admission queue. The source does not define any of these as a zero-latency or unlimited-resource guarantee. [T L174-213, L398, L755-813, L892-943]

5 Records and immutable meaning

A payload travels with the information needed to interpret it

Retain lnd_record_envelope_v1 and unchanged legacy payloads. tomoss_execution_v1 adds plan/state identity; its explicit successor tomoss_execution_v2 adds the ECS world contract in Section 43. Incompatible payload changes require a new payload profile; metadata never silently changes integer_bits_v1. [S §§7-8]

Record group	Required content
Identity and schema	Record, source, channel and dataset IDs; envelope and payload profile versions; byte length; encoding; content digest.
Observation context	Acquisition time or explicit absence; arrival sequence; clock and time scale; exposure interval where relevant.
Spatial and physical context	Coordinate frame, axis order, handedness, datum/epoch, height kind, quantity and units.
Quality and calibration	Validity, orientation-defined and applicable quality flags; calibration version; uncertainty model or explicit unavailable state.
Evidence and provenance	Source class, source span/reference, predecessor and input digests, operator version, model and parameter identities.
Execution identity	Instance ID, base state generation, descriptor digest, registry epoch, plan digest, job key and numerical/backend profile.
Result semantics	Output schema; exactness class; rounding and interpolation policy; certificate or declared error estimate.
Delivery record	Separate immutable outbox event, attempt and acknowledgement records linked to the committed result.

Canonical representation

Preserve raw source bytes. Canonical serialization has a versioned grammar. Observational completion timestamps stay outside numerical replay comparisons; acquisition times remain when they affect meaning. Retained words use binary strings with declared width/signedness; rationals keep a signed numerator and positive denominator. Null requires schema permission.

The descriptor is a separate control artifact with typed strings, enums, arrays and version references. Section 48 selects tomoss_c14n_jcs_v1 for new control records while preserving legacy payload bytes. Stable plan and cache digests require the declared canonicalization profile.

A digest detects disagreement with expected bytes. It does not, by itself, prove a person's identity or authenticate a sender. Personal identifiers do not seed physical constants or numerical state. Corrections create derived records and preserve their predecessors. [M pp. 11-13; T L586; V]

6 Exact numerical foundation

Retain the log net D words and transport frame

Quantity	Word and semantic domain
Latitude	Signed 40 bits; -90,000,000,000 through +90,000,000,000 nanodegrees.
Longitude	Signed 40 bits; [-180,000,000,000, +180,000,000,000) nanodegrees.
Local wrapped theta	Unsigned 39 bits; 0 through 359,999,999,999 nanodegrees.
Height	Signed 64 bits in millimetres; vertical reference belongs in metadata.
Winding	Signed 32 bits; complete accumulated turns.
Derived phase	U = wN + j; derived signed 71-bit representation within the valid w and j domain.

$$N = 360000000000\,,\Delta\theta = \frac{\pi}{180000000000}\,rad$$

The preserved frame

Wire field order	Bits	Offsets
Latitude; longitude	40; 40	0-39; 40-79
Height	64	80-143
Theta; winding	39; 32	144-182; 183-214
Validity; longitude defined	1; 1	215; 216
Zero padding	7	217-223

The payload is 217 bits and the stored frame is 28 bytes. Signed words use two's complement; bytes and bits are MSB-first. Reject partial frames, nonzero padding, semantic range violations and inconsistent flags. Valid storage width alone is insufficient. [M pp. 3-6, 11; V]

Additional operational rules

At zero planar radius, orientation is undefined: keep an explicit flag and use zero theta only as a placeholder. At a geodetic pole, use the declared canonical longitude convention. Undefined longitude and undefined local orientation are different conditions. Reject a redundant phase that disagrees with winding and theta.

Exact symbolic pi means a named constant with a rational coefficient. It is not an infinite digit buffer. Off-grid angular inputs fail unless a separate, recorded quantization operation is requested. Geodetic-to-ECEF evaluation remains an optional numerical service with its own error contract.

7 Phase and genuine automorphisms

Separate accumulated motion from hexagonal symmetry

Wrapped orientation and full turns

$$j = U \bmod N,\ w = \lfloor U / N \rfloor,\ U = w N + j,\ \Phi = 2\pi U / N$$

Use Euclidean division, including negative values. U = -1 gives winding -1 and theta N - 1. A phase advance adds to U and checks the resulting winding range. A wrapped orientation action changes direction without implicitly changing accumulated turns.

D6 on orientation

$$T = N/6,\ s = (-1)^b,\ g_{k,b}(j) = (s\,j + k\,T) \bmod N$$

$$(k,b) \circ (\ell,c) = \left(\left(k + (-1)^b \ell \right) \bmod 6,\ b \oplus c \right)$$

Here k ranges from 0 to 5 and b is a reflection bit. Composition applies the right-hand action first. The twelve actions preserve the declared hexagonal structure; six 60-degree rotations restore orientation, while six 60-degree phase advances add a full turn. [M p. 7]

Integer axial coordinates and array indices

$$R(q_a, r_a) = (-r_a, q_a + r_a),\ S(q_a, r_a) = (q_a + r_a, -r_a)$$

$$j'_{72} = (s\,j_{72} + 12\,k) \bmod 72$$

The axial quadratic form q_a² + q_a r_a + r_a² is preserved. The retained 72-column LUT and its mask satisfy their D6 action exactly. A new measured field may transform without being invariant. Vector components require their own representation transform. [M pp. 7-8; V]

An optional model symmetry requirement

$$F(g\,x, g\,u; c) = g\,F(x, u; c)$$

This is an equivariance test for a model whose geometry, loads, parameters and boundary conditions admit the chosen action. It is a proposed test, not an automatic property of every physical module. Degradation, clipping and state merging are commonly noninvertible; calling them feedback does not make them automorphisms. [T L177, L537; engineering correction]

8 Numerical lookup and logarithmic codes

Preserve the retained table and distinguish decoding from reconstruction

Retained property	Value or policy
LUT profile	LND_HEX_Q16_BITS_V1
Shape and mask	17 radial rows by 72 angular columns; 1,224 positions; 1,158 valid and 66 invalid.
Signal and mask packing	16 signal bitplanes: 2,448 bytes; validity mask: 153 bytes.
Sampling	5-degree angular grid; first ring at 0.5 mm; separate center; 0 < r < 0.5 mm is unrepresented.
Default access	Exact node addressing. Check index domain and mask before decoding. Reject negative indices and boolean arguments.

Retained Q16 compatibility mapping

$$L = \ln\left(1 + s_{max}/s_0\right), q = \left\lfloor 65535\,\frac{\ln\left(1+s/s_0\right)}{L} + \frac{1}{2} \right\rfloor$$

$$\hat{s}/s_0 = 16^{q/65535} - 1 \text{ when } s_{max} = 15\,s_0$$

Require compatible positive scales and a nonnegative in-range signal. The legacy profile is a 16-bit quantized signal mapping, not a six-bit logarithmic exponent. Decode before physical averaging, interpolation and exposure integration. Its inverse recovers the decoded code value; quantization prevents recovery of an arbitrary original signal. [M pp. 8, 15, 44]

Certified integer evaluation

$$q/65535 = p/d,\ A = S^d 16^p,\ n^d \le A < (n+1)^d$$

$$\frac{n-S}{S} \le \hat{s}/s_0 < \frac{n+1-S}{S}$$

For positive integer S, the existing interval evaluator returns rational bounds and a certificate. An exact power equality yields an exact endpoint. Precision, integer size and cancellation budgets are required. The proposed symbolic service returns the expression without pretending it has already been numerically evaluated. [M p. 10; V]

Keep exact_log2_levels_64 separate: v_k = 2^k - 1 for 0 ≤ k < 64. Its default encoder accepts exact levels only. Optional rounding is named. Interpolation or a new mask creates a separately versioned dataset/profile; missing nodes and the inner radial gap are never silently filled.

9 Operator registry and numerical policies

Every executable operation has a complete contract

An operator entry binds an ID and immutable implementation version to input/output schemas, units, domains, numerical policy, resource class, capabilities, side effects and error codes. Registry entries are validated before a descriptor can reference them. A description of an operator is not executable authority.

Operator family	Required contract
Validate and canonicalize	Preserve raw data; reject invalid domains; define seams, poles, flags and exact parsing.
Phase and D6	Name wrapped versus accumulated action; check winding overflow; identify the preserved structure.
Numerical LUT and decoding	Validate indices and masks; bind table digest, scale and exact/interval profile.
Pack and transform layout	Preserve complete typed fields, order and padding; state exactly what an inverse recovers.
Model step and guard	Specify base state, timestep, parameters, solver, units, tolerances and possible unknown guard result.
Render and export	Read committed results; declare lossy conversion, pixel/texture policy and invalid-value display.
Device or network output	Require a capability and stable intent ID; state acknowledgement and retry semantics.

Numerical policy is part of the result identity

EXACT_INT and EXACT_RATIONAL preserve arithmetic on their declared domains. SYMBOLIC preserves a typed expression. CERTIFIED_INTERVAL encloses the mathematical quantity under the stated certificate. APPROXIMATE records precision, rounding, method and applicable error estimate. An interval around a model calculation does not automatically enclose the unknown physical truth.

Deterministic execution has a defined scope

The strict replay profile freezes input bytes and interpretation, implementation and backend versions, configuration, reduction order, random seeds if used, and recorded external observations. Wall-clock completion times and device scheduling are observations, not canonical numerical outputs. Approximate backends can use tolerance-based conformance only under a separately named profile.

Fixed-array indexing may have constant algorithmic cost under a RAM model. Validation, hashing, integer-root evaluation, I/O and model solves have their own costs. TOMOSS.A.D. does not infer constant latency from the word LUT. [M pp. 10-12, 26; T L713-754; S §15]

10 The state descriptor table

LUT(sS) selects behavior through typed versioned records

Define sS as the discrete state of one model instance. The lookup key also includes a model/profile version and a backend capability profile. Global configuration epoch, job lifecycle and model state are separate namespaces. One instance entering a new state cannot silently change another instance. This scope is a proposed engineering interpretation of T L755.

Descriptor field	Meaning and validation
Identity	descriptor_id, schema_version, descriptor_version and content digest.
Selection key	model_profile, state_code and compatible backend_profile; exactly one matching row.
Components	Registered component IDs and exact versions; provided and required interfaces.
Bindings	Typed operator bindings, input/output profiles, parameter and dataset references.
Graph	Named ports and dependency edges; cycles require an explicit state/time boundary.
State contract	State schema, checkpoint format, ownership, migration policy and allowed transition references.
Execution policy	Numerical profile, scheduling mode, determinism class and resource limits.
Capabilities and outputs	Required capabilities, permitted effects and observer/event schemas.

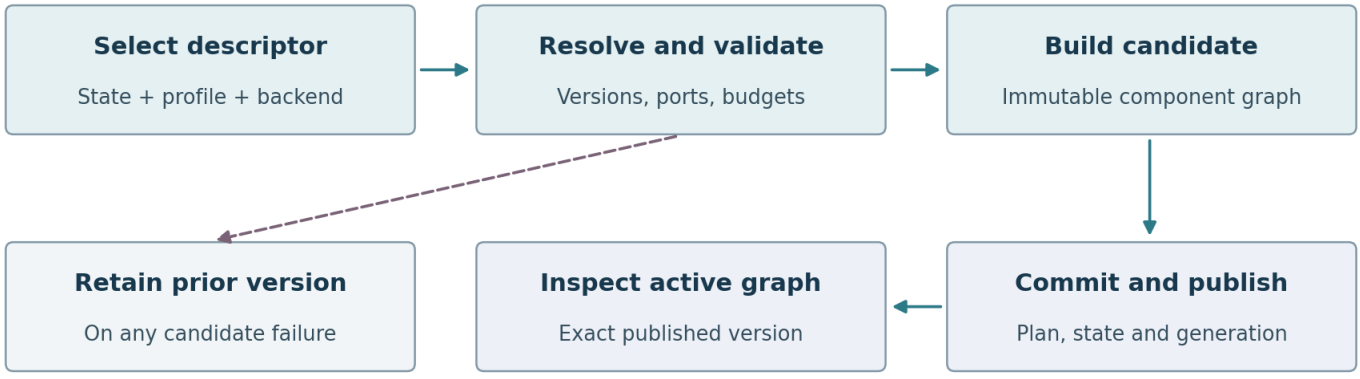
Five table roles remain separate

Typed table	Question answered
Numerical LUT	What value exists at this validated numerical index?
State descriptor LUT	Which compatible runtime components are selected for this state?
Transition table	Which next state follows from this event and evaluated guard?
Parameter table	Which measured, calibrated or illustrative parameters apply?
Rendering table	How is a committed quantity mapped into a declared visual representation?

Unknown keys, ambiguous matches, missing versions and incompatible schemas fail explicitly. No raw address, arbitrary payload reinterpretation or out-of-bounds memory operation is a valid descriptor binding. A failed candidate leaves the previous published instance usable. [T L755-890; S §§13, 15]

11 Runtime graph assembly

The diagram is derived from the instantiated graph



Old jobs retain their pinned graph until their work and device use end

Plan construction and inspection. Rejected candidates never replace the published instance. A successful graph records actual component and interface bindings.

Build and validate

The plan compiler loads one immutable registry/configuration snapshot, resolves the exact descriptor row and freezes all referenced versions. It checks schemas, units, numerical compatibility, capabilities, ownership and resource limits. It validates an acyclic within-step dependency graph; feedback crosses an explicit checkpoint or timestep boundary.

The factory instantiates only registered compatible implementations. It binds implementation-module lifetimes and immutable datasets to the plan. Construction produces either a complete candidate plus validation report or a typed error. No worker can observe a half-built graph.

Inspect what is active

The runtime diagram MUST be generated from the published plan's component graph. Show instance ID, state code, registry epoch, plan digest, component versions, ports, dependencies and backend. Distinguish required interfaces from instantiated providers. A diagram is an inspection view of that version, not a replacement for the descriptor or evidence of successful execution.

Stable interfaces permit dynamic composition

C++ classes, ABI boundaries or other language types may be defined at build time. The active composition, strategy selection and resulting diagram form at runtime. This satisfies the explicit source request without adopting the generated answer's claims about arbitrary vtable patching. OOB remains an unresolved label. [T L755-813]

A model state transition is valid only if its target descriptor and compatible state schema can be prepared. If migration is required, run a named migration under its own input/output checks. Never reinterpret a live buffer as a new state type.

12 Publication and state transitions

Durability and atomic visibility are separate steps

Each instance controller owns a published snapshot containing its plan, state-record identity, ECS world reference and increasing generation. The initial design has one state-changing owner per instance and one configuration publisher. Independent instances and pure computations can run concurrently.

Phase	Required action
Prepare	Build the next immutable plan and proposed next-state record from the pinned base generation. Validate all dependencies and migration output.
Check	Confirm the base generation is still current, the transition is allowed, capabilities remain valid and resource budgets permit old and new versions to coexist.
Commit	Atomically persist the new plan identity, next state, job result, terminal disposition, activation record and required outbox entries.
Publish	Update the instance's in-memory atomic snapshot pointer only after durable commit. This is the visibility boundary for new admission.
Retire	Retain old snapshots, modules and datasets while jobs or device submissions still reference them. Reclaim only when all relevant lifetimes end.

Active and queued work

An admitted job pins a complete snapshot and never silently adopts a newer plan. The instance owner admits at most one unresolved state-changing attempt. A successor binds its input after the preceding attempt terminates and any committed activation has been published. Configuration changes affecting the instance pass through the same owner. Other intents remain in a bounded pending queue; pure jobs may share immutable snapshots.

Crash and conflict behavior

If a crash follows durable activation but precedes pointer publication, recovery reconstructs the committed snapshot. State-changing commits require the expected base generation; stale attempts receive state_conflict. Pure snapshot results may commit after a newer state activates, retaining their original generation. A future multi-writer implementation requires version-checked commit semantics and a matching publication protocol.

Rollback means a new recorded decision

A replacement can select a previous compatible configuration under a new generation. It cannot erase committed results or undo an external action. Missing old artifacts cause recovery failure with a named dependency, never substitution of the newest version. [Proposed protocol; T L895-907; S §§4-7]

13 Concurrent snapshot execution

Ownership makes the lifetime of a published plan explicit

A proposed C++20 implementation uses an atomic shared pointer per instance. Plan, model state and ECS world are coherent together. All reachable data must be immutable; const on the outer object is insufficient. This is an interface sketch, not a compiled implementation. Section 50 supplies the admission and publication gates.

```
struct PublishedInstance {
  std::shared_ptr<const RuntimePlan> plan;
  StateRef state;
  WorldSnapshotRef world;
  Generation generation;
};
std::atomic<std::shared_ptr<const PublishedInstance>> active;

// Within the admission protocol; pin once.
auto pinned = active.load(std::memory_order_acquire);
auto job = validate_and_freeze(request, pinned);
admit_durably_with_capacity(job);
q0.notify_ready(job.id);

// After durable activation, while mutation admission is gated.
active.store(candidate, std::memory_order_release);
```

Admission requires a ready service and initialized snapshot. Reserve capacity before durable acceptance. The durable READY registry is authoritative; Q0 notification is idempotent, so a crash or notification failure cannot lose accepted work.

A release store and an acquire load that observes it provide the required CPU publication ordering. Atomic shared-pointer ownership protects the acquired object's lifetime, subject to complete ownership of what it references. The C++ specialization does not guarantee lock freedom. Release ordering does not write a journal to disk or complete GPU work. [X1-X3]

Shared object	Ownership or synchronization policy
Plan and source records	Immutable; keep all modules, function-pointer owners and datasets alive through their users.
Working model state	Owned by one state-changing job per instance; pure child tasks receive bounded immutable inputs or private output buffers.
Registry, journal and cache	Versioned registry; transactional journal; immutable cache values with an explicit concurrent map policy.
Observer queues	Bounded asynchronous outbox delivery; callbacks run outside publisher locks.
GPU resources	Retain until the relevant completion fence or documented device-loss cleanup; cancellation is not completion.

RCU is a possible later implementation with a real reader/grace-period reclamation protocol. It is not the name for any copy-and-swap pattern. The supplied Python example does not implement its claimed CAS, epochs or asynchronous observers. [T L946-1071; X4, X15]

14 Jobs and deterministic scheduling

Q0 owns admitted work and records every terminal outcome

Job state	Allowed outcome
RECEIVED	Capture request intent. Resolve dependencies and validate into READY, or terminate as REJECTED/CANCELLED.
READY	Persist frozen snapshot, inputs, job key and admission sequence. Enter Q0; start RUNNING or cancel.
RUNNING	Execute within budget. Produce PREPARED, FAILED or CANCELLED.
PREPARED	Validate output; state-changing attempts also validate the expected base generation. Commit or record failure/cancellation before commit wins.
COMMITTED	Authoritative result and outbox exist. Delivery proceeds without recomputation.
DELIVERY_COMPLETE	All required destinations acknowledged or reached an explicit terminal disposition under the delivery policy.

FAILED, REJECTED and CANCELLED retain typed causes. A retry is a linked attempt. Cancellation that loses to commit reports too_late. Every accepted attempt has a terminal sequence record. Version 0.2 names delivery completion explicitly; snapshot publication is a separate event, not a delivery acknowledgement.

Reference and concurrent modes

REFERENCE uses one worker and stable admission order. PARALLEL_REPLAY executes independent pure work concurrently but fixes dependency ordering, reduction ordering and canonical terminal commitment order. Every sequence position ends with a result or failure/cancellation record so one failed job cannot leave a permanent gap. The waiting-result buffer is bounded.

THROUGHPUT is an optional separately named mode that may commit independent jobs by completion order. It still preserves per-instance state order and records the actual event order. Approximate GPU work uses a declared conformance tolerance unless bitwise replay has been demonstrated for that backend.

Three independent state machines

Runtime lifecycle: STARTING, READY, DRAINING, STOPPED, FAULTED. Job lifecycle: the states above. Model lifecycle: a profile-specific set such as STABLE, BOUNDARY and ONSET_REACHED. A model's terminal state does not halt TOMOSS.A.D. or discard its result.

Q0 has job-count and byte limits. Admission acknowledgement follows durable acceptance. Full queues return queue_full and apply declared backpressure. Acquisition time, arrival order, admission order and publication order remain separate. Exact numerical replay excludes observational wall-clock timestamps. [S §§4-6; T L892-1071; proposed concurrent extension]

15 Five selected design patterns

Concrete roles within the runtime architecture

The attachment selects these five patterns in its generated response to BIG 5. They are adopted here as a project design set, not presented as a universal canonical list. Patterns organize responsibilities; the contracts in Sections 10-14 establish correctness. [T L909-943]

Pattern	TOMOSS.A.D. role	Required boundary
Abstract Factory	Construct a compatible component family from a validated state descriptor.	Resolve exact versions and reject incomplete or incompatible construction.
Strategy	Bind an operator implementation behind a stable typed interface.	Declare arithmetic, error, cost and capability differences; preserve output schema.
State	Select legal behavior and transitions for an instance's discrete model mode.	Keep runtime, job and model state separate; specify total guarded transition behavior.
Observer	Deliver committed changes to views, telemetry and other subscribers.	Use bounded outbox queues, stable event IDs and isolated callback failures.
Prototype	Reuse an approved immutable configuration template when constructing candidates.	Clone configuration intentionally; never assume a deep copy proves isolation or bounded latency.

A state change in operation

The State policy evaluates a guard against a validated candidate model result derived from the pinned committed base. The Abstract Factory resolves the target descriptor and binds its Strategies; Prototype reuse can supply immutable configuration. The transition becomes authoritative when result, next state and activation commit together. Observer delivery follows commit. Existing jobs keep their previous snapshot until completion.

What is shared and what is copied

Large immutable datasets and implementation modules may be shared by ownership. Per-job scratch memory and mutable simulation state are privately owned. State migration creates a new record. Copying an object graph without checking aliases, device handles, callbacks and module ownership does not establish a safe runtime plan.

Avoid accidental coupling

A renderer subscribes to committed output; it does not control the numerical solver through a texture alpha channel. A state table selects an allowed operator; it does not bypass domain validation. An AI-generated descriptor is a proposal until the same registry, capability and acceptance gates approve it.

16 Evidence predicates and entropy

An observation and a belief are different records

Retain the source's useful distinction between a model's hidden state, a recorded observation and an observer's decision. Formalize it without personal identity or theological predicates. A record stores a proposition, its supporting evidence, any inference rule and the policy that acts on the evaluated result. [T L220-325, L586]

Evidence status	Proposed representation and meaning
UNKNOWN	No accepted support for the proposition or its negation.
SUPPORTED_TRUE	Accepted positive support; no accepted negative support.
SUPPORTED_FALSE	Accepted negative support; no accepted positive support.
CONFLICT	Accepted positive and negative support coexist and require a declared resolution policy.

Two support bits can encode these four states, with provenance attached to each support. They describe evidence, not universal truth. Numerical validity, measurement quality and execution authority remain separate fields. Not knowing a proposition does not establish its negation.

Guards must be complete

A transition table names its event schema and guard. Guards are disjoint and exhaustive under the profile, including equality, unknown and invalid cases. A guard requiring calibrated input returns unavailable when that calibration is missing. Unknown does not default to the stable row. An interval that crosses a threshold produces BOUNDARY or a documented refinement request.

Entropy has a declared mathematical object

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p), 0\log_2 0 := 0$$

This Shannon entropy is measured in bits for a declared binary probability distribution. It is one bit at p = 1/2 and zero at p = 0 or 1. A confident but wrong distribution can have zero entropy. The metric does not establish truth or observer omniscience. [X10]

Thermodynamic entropy has units J/K and belongs to a physical state/boundary model. Damage is a separately defined structural variable. No universal inverse law among damage, thermodynamic entropy and observer uncertainty is adopted. The source's divergent collapse values and numerical entropy table are unsupported illustrations. [T L326-393; X9]

17 Source and domain adapters

Connect data sources through explicit interpretation contracts

Adapter	Captured meaning and required boundary
Documents and code	Source bytes, document spans, repository revision, language/parser version and syntax-tree relationships. Text positions are not physical coordinates.
Sensors and time series	Actual channel, units/raw counts, acquisition clock, sampling/exposure interval, quality and calibration. Gaps and saturation remain explicit.
Geospatial data	Dataset revision, license/provenance, CRS, axis order, coordinate epoch, vertical reference, resolution, footprint and no-data policy.
GNSS and navigation	Receiver observations, time/reference frame, quality and available receiver metadata. Navigation observations do not provide an application uplink.
Network transport	Configured endpoint and protocol, authentication capability, delivery IDs, sequence policy, reconnect behavior and availability.
Material and optical experiments	Specimen/batch ID, preparation, geometry, conditions, calibrated exposure history and measured response.
Model output	Model, parameter and solver versions, assumptions, input uncertainty and numerical error profile.
Rendering or export	Committed record identity, display/export version and any irreversible conversions.

Project extension slots

The project context mentions Earth Engine, AHN3, phone sensors, time-series transport and a tiled GPU view. These can occupy the adapter interfaces above. The attachments do not establish access to those services, available credentials, the user's live sensor inventory or a working integration. Implementation begins with an exact source/capability manifest, then a small recorded fixture.

Exposure and material interpretation

$$H_{rad}(X) = \int I(X, t)\, dt$$

Irradiance and exposure require compatible units and the material point's trajectory. Raw pixel counts, an analog logarithmic amplifier, a digital log encoder and a calibrated physical signal have different transfers. Decode and calibrate before physical arithmetic. Material conversion, stiffness, crystallinity and strength need their own evidence; an optical pattern alone does not establish them. [M pp. 20-24; S §§16-19]

Optional special-function or communications modules retain distinct namespaces: Fresnel reflectance, Fresnel diffraction integrals, polar coordinates and polar error-correcting codes are not interchangeable operations.

18 AI compilation and bounded memory

Generated proposals enter the same validated runtime

The AI layer may interpret source material, draft descriptors, propose code, retrieve context and explain committed results. A generated proposal has its own evidence class and provenance. It cannot directly publish executable behavior or acquire a device capability. The attachment supplies no integrated AI compiler or model inventory.

Stage	Proposed contract
Model and dataset registry	Exact model/dataset identifier and revision, digest, format, license, tokenizer/schema, runtime and capability requirements. No inferred private account inventory.
Plan compilation	Parse descriptor, resolve versions, type-check ports/units, validate graph and budgets, select supported backend and emit plan or rejection.
Optional native compilation	Source revision -> typed intermediate form -> build artifact -> validation -> compatibility tests -> reviewed registry entry. Record compiler, flags and dependencies.
Runtime activation	Use the same prepare, commit and publication protocol as any other plan. Code generation is not activation.
Context retrieval	Index immutable source chunks with source spans and versions; retrieve under a token and latency budget; expose selected evidence.
Cache	Memoize pure results by all meaning-bearing inputs, profiles, versions, parameters, calibration and relevant backend policy.

Plan compilation and JIT are separate choices

The initial implementation can construct runtime plans from packaged operators without generating machine code. A later C++ JIT profile must define supported host platforms, compiler availability, isolation, ABI, artifact verification, cancellation and rollback. Mobile deployment starts with packaged vetted components; arbitrary downloaded native code is not assumed. [X13]

Finite context with external memory

A persistent source store can exceed the model's active context. Retrieval, summaries and checkpoints make selected material available within a finite context window. They do not create infinite attention or remove RAM, storage, compute and latency limits. A run records which source chunks were actually presented to the model.

Cache eviction cannot erase the authoritative journal, source evidence or committed results. Content-addressed reuse is valid only for pure computations with complete keys. AI outputs and approximate kernels retain their declared reproducibility limits; changing the model or tokenizer invalidates affected cache identities. [Project-context extension; proposed architecture]

19 Optional mobile deployment

A capability profile for the project phone target

The POCO X7 Pro 12 GB target comes from the supplied project context, not the new text attachment. Treat it as an optional deployment profile. Published product-family specifications do not establish the user's exact storage variant, current installed build or available app memory.

Published property	Architectural implication
Dimensity 8400-Ultra; Mali-G720	Select a tested ARM/native backend and query actual graphics-driver capabilities.
12 GB LPDDR5X variants; UFS 4.0	Device totals are capacity context. Configure application budgets from observed availability and host limits.
5G support	Link availability depends on region/operator; offline capture and replay remain useful.
Published HyperOS 2	Record the actual installed OS/build, ABI and application lifecycle behavior at deployment.

These specifications are from the manufacturer. No TOMOSS.A.D. benchmark, sustained throughput, model-size limit or energy result has been demonstrated on the device. [X11]

Host integration rules

Run the runtime inside the Android application boundary. Query permissions and capabilities; denied or revoked access returns a typed unavailable outcome. Choose background-work mechanisms for the required lifecycle, checkpoint long work and keep computation off the UI thread. A source request cannot bypass the host permission model. [X12, X14]

Android can limit application memory and terminate cached processes. Persist authoritative state, bound caches and treat restart as normal recovery. Do not interpret 12 GB as an unrestricted heap. Packaged, vetted operators selected by data are the initial code profile; dynamic-code loading needs a separately justified deployment design. [X13, X16]

GPU ownership and validation

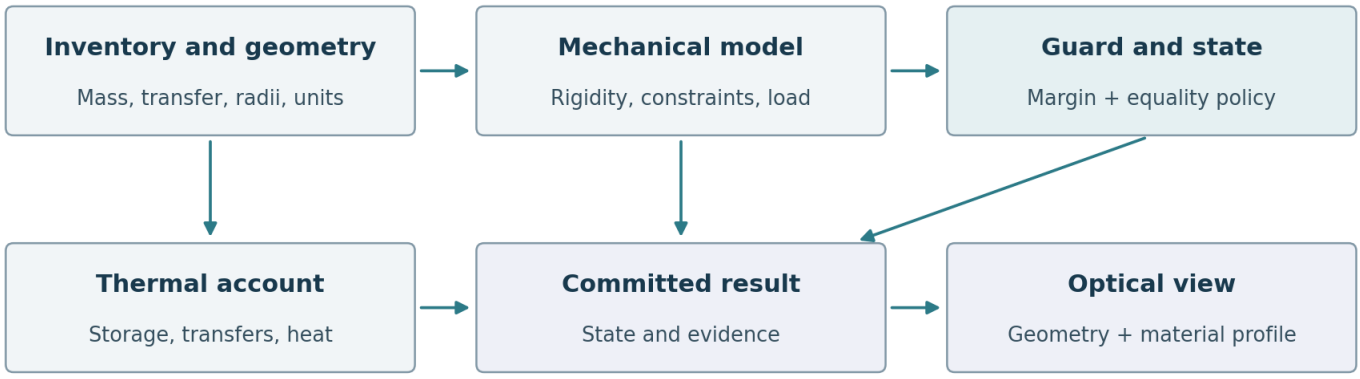
Query the actual driver, API version, features, formats and memory limits. Shared physical memory does not eliminate cache, visibility or synchronization requirements. CPU atomic publication and GPU command completion are distinct. Retain submitted resources through the graphics API's completion mechanism. [X15]

Use the CPU exact reference as the initial numerical authority. Graphics outputs are derived display records unless an accelerated numerical profile passes its own conformance checks. Device sensors, installed apps and external repositories become available only through observed, authorized interfaces.

20 Reference workload architecture

Material degradation becomes a typed hybrid model

The new source motivates a useful demonstration: material is removed or converted, geometry and stiffness evolve, a stability margin changes, and a renderer presents the committed state. The attachment supplies no measured specimen, complete parameter set or validated historical reconstruction. This workload is illustrative until calibrated. [T L586-712]



Optional synthetic workload; calibration and model validation remain separate

Optional workload graph. Mechanics, thermal accounting and rendering consume declared state. A visual change does not feed back into physical parameters unless an explicit model defines that coupling.

State or result	Required fields
Material inventory	Reservoir/species masses or amounts, reaction/transfer definitions, geometry and units.
Mechanical state	Supports, load, constitutive law, geometry, displacement/velocity and solver state.
Thermal state	Energy inventory, thermal capacity model, boundary heat/work and transfer terms.
Optical state	Surface geometry/normals and separately defined optical material parameters.
Discrete mode sS	A model classification from exhaustive guards; distinct from job/runtime state.
Observation state	Recorded measurements, inferred predicates, uncertainty and action policy.

Write the hybrid state as x_n = (sS_n, z_n), where z_n contains the typed numerical state. The update F consumes a fixed event/timestep and parameter snapshot. At a transition, the controller validates the target descriptor and any state migration before activation.

Each module declares what is assumed and what is measured. A model can be deterministic yet inaccurate. Exact geometry or arithmetic does not validate a constitutive law, reaction rate, boundary condition or failure prediction.

21 Material geometry and conservation

Define the boundary before writing a degradation law

Mass accounting

A mass-closed enclosure can contain solid material, organisms, liquid, gas and products. The eroding support is an open subsystem: removed material enters another enclosure inventory. Mass crossing the outer boundary requires an open-boundary profile. A closed boundary can exchange heat or work; an isolated boundary excludes those exchanges as well. [T L589-620; X8-X9]

$$m_{total} = \sum_i m_i \, , \dot{m}_s = - \int_{\Omega_s} R_m \, dV$$

R_m is a nonnegative mass-removal rate density in kg/(m³ s). A cross-section integral instead gives loss per beam length. For chemical conversion, use a named stoichiometric matrix and require element/charge balances; stoichiometry does not specify kinetics. The source's species-dependent rates are uncalibrated.

Annular idealization

$$A = \pi \left(R_o^2 - R_i^2 \right) , I = \frac{\pi}{4} \left(R_o^4 - R_i^4 \right) , 0 \le R_i < R_o$$

$$\frac{I}{I_0} = 1 - \left(\frac{R_i}{R_o} \right)^4 , \dot{I} = - \pi \, R_i^3 \dot{R}_i \left(R_o \, \text{fixed} \right)$$

R_o and R_i are radii in metres; A has units m² and I has units m⁴. The denominator is 64 only when using diameters. Removing central material does not imply an exponential loss of I. Density multiplied into the area integral changes the quantity and cannot silently replace geometric I in EI. [T L615-620; direct geometric derivation]

Heterogeneous material

$$D_y = \int_A E \left(x \, , y \, , t \right) \left(x - x_N \right)^2 d \, A$$

D_y is bending rigidity about a properly defined neutral axis. The profile must distinguish geometric removal, density evolution and modulus damage to avoid counting the same loss repeatedly. A radial law with 1/r requires an origin policy and dimensional parameters; the source's singular law is not adopted as a calibrated default.

Acceptance requires nonnegative inventories, valid geometry, balanced transfers and explicitly declared residual tolerances. A geometric center may be represented separately rather than assigned an infinite log-radius or a fabricated physical value.

22 Structural mechanics and onset

A model margin is evaluated under declared assumptions

Slender elastic column profile

$$P_{cr} = \frac{\pi^2 E I}{\left(k_{eff} L\right)^2}, m_P = P_{cr} - P$$

The reference workload evaluates quasi-static snapshots. E is Young's modulus, I the area moment, L the length and k_eff the support-dependent effective-length factor. This idealized onset criterion requires slender elastic geometry and conservative loading assumptions. It does not predict fragmentation or a postbuckling trajectory. Instability can occur at positive I. [X6]

Finite element branch

$$M\ddot{u} + C\dot{u} + f_{int}(u, z) = f_{ext}, K_t = \partial\left(f_{int} - f_{ext}\right)/\partial u$$

The displayed dynamic branch assumes fixed mass and a declared configuration; material-removal dynamics also needs a momentum-transport model. Declare mesh, DOFs, constraints, material/load laws, integrator, solver and convergence criteria. Section 35 retains a limited Hermite beam element; it does not cover contact, plasticity, fracture or all dynamic instabilities.

$$\left(K_c - \lambda K_{G0,c}\right)\phi = 0$$

For a conservative linearized buckling model, use constrained elastic stiffness K_c and a positive compression geometric-stiffness convention K_G0,c. The smallest relevant positive load multiplier estimates onset. A differently signed geometric matrix requires a corresponding sign change. Record residuals and the load pattern. [X7]

Guard semantics

Computed margin or condition	Model classification
Certified lower margin > 0	POSITIVE_MARGIN within the declared model and included uncertainty.
Certified upper margin < 0	ONSET_EXCEEDED within that model.
Interval includes zero	BOUNDARY; refine, pause or follow a declared unresolved policy.
Invalid constraints or failed solver	MODEL_UNAVAILABLE; no normal-state fallback.

For a constrained symmetric conservative tangent system, positive definiteness can be assessed through a consistent scaled basis and eigenvalue/residual tolerances. A determinant sign is insufficient: diag(-1,-1,1) has positive determinant and negative directions. Nonconservative, gyroscopic or damped dynamic stability needs the appropriate dynamic eigenproblem. [T L667-706; engineering correction]

23 Energy heat and physical entropy

Use a consistent inventory and dimensional balance

Energy boundary

$$\frac{d\,E_{total}}{d\,t}=\dot{Q}_{in}-\dot{W}_{out},E_{total}=U+E_{kin}+E_{pot}$$

This closed-system convention counts heat into the boundary and work out. U consistently includes thermal, chemical and stored elastic energy. For an isolated system both boundary transfers vanish. Internal dissipation redistributes energy. Tracking only the degrading support requires an open-subsystem balance with mass/enthalpy transfer. [X8-X9]

Power and heat storage

$$P_{diss}=\mu F_n v_{slip}+\int_{\Omega}\beta\,\sigma:\dot{\varepsilon}_{inel}\,dV$$

The friction term has units W. The stress-strain-rate product is W/m³ and must be integrated over volume. Beta is a declared fraction converted to heat. Terms may be added only if they represent distinct dissipation mechanisms; do not count a contact contribution twice. [Correction of T L686-696]

$$C_{th}\dot{T}=P_{heat}+\dot{Q}_{boundary}-P_{other}$$

This proposed lumped approximation uses constant C_th in J/K. P_heat is allocated dissipation plus chemical heat release, counted once and debited from the corresponding stored-energy inventories. P_other excludes transfers already included in net boundary heat. Changing mass/composition requires its storage and flux terms. No temperature spike follows from the source alone.

Entropy and observability

$$\Delta S_{isolated}\geq 0,\left[S_{therm}\right]=J/K$$

This thermodynamic statement concerns the complete isolated system. It neither sets the entropy change of a mass-losing subsystem nor determines an observer's Shannon entropy. Structural damage, stored energy, generated heat and information uncertainty are separate state variables unless a tested constitutive or observation model links them. [X9]

Validation boundary

Check dimensions, signs, inventories and energy residuals. Benchmarking checks numerical solutions against known problems. Calibration estimates sensor/model parameters from reference measurements. Validation assesses predictions against relevant observations, preferably independent of calibration. These establish different things.

24 Graphics and log polar views

Render committed model outputs through a declared display profile

Coordinate mapping

$$u = \frac{\ln\left(r / r_{min}\right)}{\ln\left(r_{max} / r_{min}\right)}, v = \frac{\theta \, mod \, 2\pi}{2\pi}, 0 < r_{min} < r_{max}$$

This mapping is defined on the stated annulus. The center has a separate representation and orientation-defined flag. Preserve the numerical LUT's unrepresented inner region unless a new sampling profile supplies data. A log-polar coordinate transform is not automatically compression or physical scale invariance. [M pp. 16-18; T L426-472]

Fresnel approximation and material parameters

$$F\left(\vartheta\right) \approx F_0 + \left(1 - F_0\right)\left(1 - \cos\vartheta\right)^5$$

A renderer must define the angle convention and approximation profile. For a microfacet reflection model, Fresnel combines with a facet distribution, visibility/masking terms and directional normalization. The source shader supplies an illustrative Fresnel display, not a complete coupled mechanical and optical solver. [X5]

Deformation can change normals, visibility and roughness. It does not automatically change refractive index or F0. A material-state-dependent optical change requires a separate parameter or constitutive model. The source's F0 table is illustrative and uncalibrated. [T L416-437, L622-637]

GPU data class	Required representation
Continuous display fields	Named channels, units, precision, filter mode and invalid mask. RGBA32F means four 32-bit float components, 128 bits per texel before layout overhead.
Categorical state IDs	Integer buffer/texture or exact nearest access; no interpolation of state meaning.
Exact packed numerical words	Explicit integer/limb layout or a declared lossy display conversion; no silent float32 replacement.
Opacity and color	Physical/display alpha is separate from state code. Define linear/color-space conversion and tone mapping.

Normals used in the lighting calculation must be the normals actually derived by the chosen model. The supplied shader computes an unused perturbed normal and unused inertia; those lines do not establish coupling. GPU fences/barriers and resource lifetimes belong to the backend contract. A visually plausible frame is not a physics validation result. [T L455-471; X15]

25 Worked state transition

Exact thresholds produce an inspectable runtime change

This synthetic example demonstrates scheduling and state selection. It is not a calibrated material law. Start with exact integrity d_0 = 1, subtract 1/20 per tick and use a strict threshold 3/20. Keep the state in exact rational or equivalent integer units.

$$d_n = max(0, 1 - n/20), trigger_n = [d_n < 3/20]$$

Tick	Exact integrity	Guard and descriptor
16	1/5	Above threshold; stable_demo descriptor.
17	3/20	Equal to threshold; boundary_demo descriptor under this proposed three-way policy.
18	1/10	Strictly below; onset_demo descriptor.

The source's repeated binary floating subtraction reaches 0.1499999999999997 at tick 17 and therefore triggers its < 0.15 test one tick earlier. That reproduced arithmetic difference shows why equality and representation belong in the model profile. It does not measure a real failure time. [T L267-306; V2]

The runtime transaction at tick 18

The controller admits the tick against the committed tick-17 snapshot. The worker computes the new value using the pinned exact policy and prepares the guard result. The factory builds onset_demo, checks its state schema and prepares a new instance snapshot. The transaction commits the result, next state, activation record and observer event. Only then is the new snapshot published.

A diagnostic job already admitted on tick 17 may finish with its original snapshot. Its result remains labeled with that state generation. A successor state-changing tick waits for tick 18 to commit and its activation to be published before binding input. The tick-18 graph shows onset_demo and its actual component versions.

A numerical-core transaction in the same architecture

$$355000000000 + 10000000000 = 1 \cdot N + 5000000000$$

The exact phase operation advances 355 degrees by 10 degrees to winding 1 and wrapped theta 5 degrees. It uses the same record, job and provenance infrastructure without requiring the material model. Q16 code 21,845 similarly decodes to 16^(1/3) - 1; a certified scale of 1,000,000 encloses it between 1.519842 and 1.519843. [M pp. 4-6, 10; V]

26 Failure and capability contracts

A failure preserves useful state and has a typed cause

Failure class	Required result
Unknown schema/profile/state	Reject before execution. Preserve the source and descriptor reference; no guessed default.
Domain error or overflow	Identify the offending value and range. Wrap, saturate or quantize only under an explicitly selected operation.
Invalid node or unsupported region	Return unavailable data. Preserve mask semantics and the distinction from zero.
Missing calibration or metadata	Store the raw record where valid; block only interpretations requiring the missing context.
Graph or capability mismatch	Reject the candidate plan, retain the current instance and report the incompatible edge/capability.
State conflict or migration failure	Keep the authoritative generation. Retry only through a declared rebinding or migration procedure.
Queue, storage or memory limit	Decline new admission. Persistence failure for accepted work pauses/faults the service; retain recoverable state and record a terminal cause when storage permits.
Precision unavailable or solver failure	Return failure or a separately permitted deferred symbolic result; never invent a numerical certificate.
Observer or delivery failure	Keep the committed result and pending outbox event; retry under bounded policy.
Device loss or acknowledgement unknown	Follow backend cleanup and external reconciliation; do not assume an action did or did not occur.

Capability records

A capability identifies an available authorized operation, scope, expiry/lifecycle and host interface. Reading a sensor, loading a dataset, computing a result and commanding an actuator are separate grants. Revocation is checked at the operation boundary; a prepared plan does not guarantee continuing device access.

Error record

Record the request/job/instance IDs, plan and state generation, stage, typed error, relevant input/descriptor reference and a useful explanation. Never replace a previous valid result with zero or a partial object. Logging itself must be bounded and must not copy credentials into provenance.

A failed update should leave the previous valid published snapshot usable unless the underlying capability has itself become invalid. Runtime recovery then reports the unavailable capability explicitly. [S §21; proposed TOMOSS.A.D. contract]

27 Persistence and recovery

A committed result survives a process restart coherently

Authoritative stores

Persist immutable source artifacts, records, descriptor/registry versions, implementation artifacts required for replay, model checkpoints, the job journal and transactional outbox. Caches and rendered views can be rebuilt. Plan digests refer to retained canonical artifacts rather than process-specific pointer addresses.

Crash window	Recovery rule
Before durable admission	No accepted execution job exists; a repeated request intent can be validated again.
After admission before execution ends	Reconstruct the frozen plan and inputs; resume a compatible checkpoint or rerun pure computation under the original profile.
During result transaction	Expose either the complete committed result/state/outbox set or none of it. Resolve using the storage transaction log.
After activation commit before pointer update	Reconstruct and publish the committed instance generation.
After commit before delivery	Dispatch the existing outbox event; do not recompute the result.
After recipient action before acknowledgement	Retry with the same event ID if the recipient deduplicates; otherwise reconcile the uncertain outcome.

Idempotency and side effects

Pure computation keys cover all input meaning, operator versions, parameters and numerical policy. A command uses a distinct intent ID because two identical payloads may represent two legitimate actions. Exactly-once local commitment does not establish exactly-once physical effect across a network.

Checkpoint and artifact compatibility

Checkpoint records identify state schema, model/solver version, step, plan and input history. A restart under a different implementation requires a supported migration or explicit fresh run. Retain old artifacts while jobs, checkpoints or retention policy need them. Missing dependencies produce recoverable, specific errors.

Storage guarantees must be chosen

Sections 48-51 select a local SQLite profile, canonical records and a complete admission/commit/publication protocol. Its crash behaviour still requires host verification. Pointer publication, content identity and backup have separate roles. Retention and restore are deployment policies with acceptance checks. [S §§4-7; proposed protocol]

28 Resource budgets and observability

Measure the complete operating path on a named workload

Budget	What must be bounded
Admission	Pending intents, admitted jobs, input bytes and completed results waiting for ordered commit.
Plans and state	Graph nodes/edges, construction time, state size, simultaneous pinned generations and migration scratch space.
Numerical work	Integer bit size, interval precision, solver iterations, timestep count and cancellation checkpoints.
AI and retrieval	Model weights, working context, retrieved token count, index/cache footprint and inference time.
GPU	Staging and device buffers, texture size, in-flight submissions and resource retirement backlog.
Persistence and delivery	Journal/outbox growth, retry attempts, retention and restore space.

A memory account includes retained versions

$$B_{peak} \geq B_{active} + B_{pinned} + B_{candidate} + B_{jobs} + B_{GPU} + B_{cache}$$

The categories must be defined to avoid double-counting shared allocations. The inequality is a planning account, not a measured device value. Slow jobs can retain old plans; reconfiguration applies backpressure when the retained-version budget is exhausted.

Operational measurements

Report queue depth, admission/rejection counts, latency distributions, throughput, peak resident memory, plan construction time, pending generations, interval widths, solver residuals, failure causes, outbox age and device loss. On mobile, include sustained thermal behavior and energy for the named test run.

Performance reports have a reproducible identity

Record device/OS/driver, software versions, backend, workload inputs, numerical policy, warm/cold state, duration and sample count. Separate pure kernel time from end-to-end time. Report observed distributions and resource limits rather than inferring real-time guarantees from design-pattern names.

The attached materials provide no integrated TOMOSS.A.D. latency, throughput or energy benchmark. Acceptance begins with correctness and bounded failure behavior, then measures implementation performance. Optional acceleration is justified by those results, not by memory type or a claim of automatic cache coherence. [T L1068-1071; X15-X16]

29 Verified numerical baseline

Evidence already reproduced in this conversation

The following results belong to the embedded log net D reference package and the earlier specification audit. They are carried into this blueprint without claiming that the new concurrent runtime has been implemented. The embedded code was inspected before execution. [V; S §§23-24]

Verification	Observed result
Bundle checks	69 of 69 passed.
Integer reference checks	46 of 46 passed.
Q16 interval checks	14 of 14 passed; 8 certificate fixtures.
Integer-level quantizers	All 65,536 nonnegative 16-bit inputs checked independently.
Retained LUT	All 1,224 positions preserved; 1,158 valid and 66 invalid.
D6 node actions	14,688 independent comparisons passed.
D6 bitplane actions	235,008 independent comparisons passed.
Regeneration	Integer packet, integer QA and Q16 QA byte-identical; rebuilt ZIP byte-identical.

The archive has 16 entries and 285,893 bytes. Its SHA-256 digest is shown below; concatenate the two printed lines. The digest identifies selected bytes, not authorship.

```
058c870f9171d12c0d298928126ec28f9
c35da643a9fb5c5c7c9e1302d7fca08
```

Known integration gaps

The existing packer can accept some semantically invalid values that the decoder rejects. A validating wrapper must reject latitude above 90 degrees, theta equal to N and inconsistent flags before packing. Redundant supplied phase must be checked rather than ignored. Legacy negative-index LUT access must be blocked. Orientation-defined metadata and runtime precision budgets must be added in the envelope/service layer.

New attachment checks

The supplied dynamic factory refers to undefined Object and raises NameError. Its concurrency sketch does not implement the claimed CAS/epochs or asynchronous callbacks. The float threshold discrepancy in Section 25 was reproduced. The three beam matrices were independently checked by exact integration of their Hermite basis. These checks diagnose source examples; they are not an integrated TOMOSS.A.D. implementation test. [T L655-665, L854, L868, L946-1071; V2]

30 Runtime acceptance gates

Observable outcomes required before implementation claims

Gate	Required observation
A01 Input boundaries	Reject invalid semantic ranges, negative indices, ambiguous state keys, missing versions and incompatible units before execution.
A02 Graph construction	An invalid edge, cycle, capability or resource requirement never replaces the active instance. Inspection matches actual bindings.
A03 Publication race	Concurrent admission/replacement binds each job to one complete generation with no mixed plan/state fields.
A04 Lifetime isolation	A long job remains valid after registry replacement; modules and inputs survive; one job cannot mutate another's state.
A05 Replay	Varied worker interleavings reproduce canonical results/certificates in PARALLEL_REPLAY and match REFERENCE.
A06 Ordered termination	Slow, failed and cancelled jobs cannot leave sequence gaps; waiting-result memory stays bounded.
A07 Cancellation race	Every attempt has one authoritative terminal outcome; a committed result survives late cancellation.
A08 Crash transaction	Injected crashes expose a complete result/state/outbox set or none; restart retains original plan/input identities.
A09 External delivery	Stable event IDs support deduplication or explicit reconciliation after an ambiguous acknowledgement.
A10 GPU retirement	Unsignalled submissions retain resources through cancellation/reconfiguration; device loss follows its cleanup path.
A11 Exhaustion	Memory, storage and queue failures preserve the last valid snapshot and produce bounded typed outcomes.
A12 Model validity	Conservation, constraints, solver convergence and uncertainty checks pass; physical claims also have relevant calibration/validation.

Run race and crash tests at the actual publication, admission, commit and delivery boundaries. Unit tests that merely echo implementation choices are insufficient. A passing numerical bundle does not waive these runtime gates. A passing runtime gate does not establish a physical material claim.

Performance targets are selected only with an explicit device/workload budget. No hard real-time or lock-free claim is permitted until the concrete implementation and relevant assumptions support it. [Proposed acceptance specification]

31 Implementation sequence

Build a small conforming runtime before expanding adapters

Milestone	Deliverable and exit condition
1 Preserve and wrap the core	Freeze existing artifacts; add strict semantic validators and typed errors; preserve byte compatibility and pass the numerical baseline plus boundary cases.
2 Implement records and journal	Canonical schemas, immutable source/results, transactional admission/commit/outbox and recovery fixtures.
3 Implement reference execution	One worker, Q0 limits, operator registry, cancellation, checkpoints and deterministic example jobs.
4 Implement LUT(sS) plans	Descriptor schema, graph validator, factory, per-instance state owner, migration contract and graph inspection.
5 Add concurrent execution	Pinned immutable snapshots, bounded worker pool, ordered replay mode, lifetime/race/crash acceptance.
6 Add the reference workload	Synthetic geometry/beam/thermal/graphics modules, independent benchmarks and explicit numerical policies.
7 Add device and AI profiles	Observed host capabilities, a small selected model/dataset manifest, supported rendering backend and measured resource budgets.
8 Calibrate and optimize	Relevant measured data, model validation and sustained target-device benchmarks before application-specific claims.

Suggested implementation boundaries

Organize packages around contracts: core/types, core/numerics, records, operators, descriptors, plans, instances, scheduler, journal, adapters, models, render and conformance. Language bindings expose the same profiles. The numerical package remains separately testable from the runtime and each domain model.

Version 0.2 implementation decisions

Section 38 selects the host/language profile, local store, canonicalization, ECS ownership and first fixture. Sections 44-55 define the ABI, compiler tiers and deployment gates. Concrete memory/queue limits remain workload and device measurements. The defaults retain one publisher, one state-changing owner per instance, packaged operators and a CPU reference path.

TOMOSS.A.D. and OOB need no invented expansion to begin implementation. Preserve the names and record any later user definition as a terminology update. The next concrete build target is milestones 1-4: a validated runtime graph executing and recording the exact examples. [Proposed roadmap]

32 Source reconciliation register

What is retained and how it enters the architecture

Source evidence	Disposition in this blueprint
M pp. 3-13 and embedded bundle	Retain exact words, phase, D6, numerical LUT, logarithmic levels, interval decoding and packed transport semantics.
M pp. 15-29 and earlier appendices	Retain typed spatial, exposure, material, communications and visualization extension points with their assumptions.
E pp. 1-8	Retain architectural vocabulary and requested workflow. Do not present generated Codex-internals claims as established implementation facts.
S §§2-7, 15, 21-25	Carry forward the record/job architecture, source provenance, validation findings and reproduced evidence; extend the initial single-worker design.
T L586	Apply the explicit scope correction: retain mechanics; exclude personal identifiers and religious/philosophical semantics from runtime models.
T L713-754	Formalize the operator inventory as typed, versioned service contracts.
T L755-813	Adopt runtime formation of the active interface/component graph from LUT(sS). Reject invented out-of-bounds execution.
T L892-943	Add concurrency; adopt the five generated-source pattern selections with explicit project scope.
T L946-1071	Salvage candidate construction and publication; replace unsupported RCU/CAS/real-time claims with an implementable protocol.
Project continuity context [P]	Keep phone, AI/model registry and geospatial adapters as optional targets, separately attributed from attachment requirements.

Source identifiers

[M] log_net_D_master.pdf, 50 pages, edition 2. [E] the supplied Google Search explanatory PDF dated 30 October 2026, 9 pages. [T] Pasted text(1).txt, the new conversation transcript. [S] LOG_NET_D_Operating_System_Specification.pdf, version 0.1, 27 pages, produced earlier in this conversation.

[V] Reproduced reference-bundle checks recorded with S. [V2] Checks of T: code defects, exact-versus-floating threshold behavior and beam-matrix algebra. Neither claims physical testing. [P] Project continuity context supplied with this request: prior POCO X7 Pro 12 GB, edge AI, Earth Engine/AHN3 and phone-telemetry targets; no additional project files were inspected.

Line references use the attachment's original line order. Apparent short user prompts and long generated responses are distinguished during synthesis; the text export has no explicit role metadata. TOMOSS.A.D. does not occur in that attachment and is supplied by the current request.

33 Corrections to generated source claims

Preserve useful intent while repairing formal errors

Source claim or defect	Adopted correction
Feedback described as automorphism	Use state update/feedback for evolution; reserve automorphism for a structure-preserving bijection such as the defined D6 action.
Runtime diagram requires vtable/OOB mutation	Materialize registered typed components and derive the inspection graph. OOB expansion remains unresolved.
Python copy/swap proves lock-free RCU	Specify real synchronization and ownership. Shared-pointer snapshots and RCU are distinct mechanisms.
Unknown state selects default behavior	Reject unknown or ambiguous selection; preserve the previous valid snapshot.
Closed means isolated	Declare the boundary and mass, heat and work exchanges separately.
Annular radius formula uses /64	Use /4 with radii; /64 with diameters. Density-weighted moments have different dimensions.
det(K-KG) <= 0 detects all failure	Apply constraints and a model-appropriate stability/eigenvalue criterion; include numerical tolerances.
Collapse creates infinite entropy/velocity	No such conclusion follows. Track physical state, damage and information entropy separately.
Heat formula adds W to W/m³	Integrate volumetric dissipation and use a consistent energy balance without double counting.
Geometry forces F0 to jump	Separate normals/geometry from optical material parameters; require a model for any coupling.
A shader demonstrates full mechanics	Unused normal/inertia variables do not establish a solver. Rendering consumes validated committed model output.
Hash or personal identifier determines physics	Keep provenance and authorization distinct from physical parameters; follow the source's explicit exclusion of personal ID.

Relevant source locations: automorphism L177/L537; radius formula L615-620; determinant and norm L667-706; thermal balance L686-696; graphics L416-472; runtime factory L755-890; concurrency L895-1071. These findings concern generated explanations and demonstrations, not a proven user implementation.

The logic/entropy corrections also prevent unsupported conclusions: one observer's missing fact does not establish a universal claim; unknown is not false; zero Shannon entropy does not imply truth. [T L240-250, L326-393]

34 Descriptor example and interface

An illustrative control record for one model state

This human-readable descriptor fragment demonstrates the proposed control schema. Referenced versions are illustrative names, not claims that those components exist. A deployable descriptor also supplies canonical artifact digests, complete budgets, bindings and validation evidence.

```
{
  "schema": "tomoss_state_descriptor_v1",
  "descriptor_id": "integrity_demo.boundary",
  "descriptor_version": "1",
  "model_profile": "integrity_demo_v1",
  "state_code": "BOUNDARY",
  "state_schema": "integrity_rational_v1",
  "backend_profile": "cpu_exact_reference_v1",
  "components": [
    {"id": "step_rational", "version": "1"},
    {"id": "guard_threshold", "version": "1"},
    {"id": "record_result", "version": "1"}
  ],
  "edges": [
    ["step_rational.out", "guard_threshold.in"],
    ["guard_threshold.out", "record_result.in"]
  ],
  "transition_table": "integrity_transitions_v1",
  "parameters": "integrity_demo_parameters_v1",
  "numerical_policy": "exact_rational_v1",
  "required_capabilities": ["compute", "result_write"],
  "budget_ref": "small_reference_job_v1",
  "migration_policy": "same_schema_only",
  "observer_event": "committed_instance_v1"
}
```

Public interface shape

BuildPlan(key, registry_snapshot, capabilities, input_manifest) returns an immutable candidate and validation report or a typed error. Admit(request, snapshot) freezes a job and persists acceptance. Execute(job) prepares output. Commit(prepared, expected_generation_if_mutating) records the outcome atomically. Generation equality is required for state changes; pure results may commit against their retained older snapshot. Inspect(plan_digest) reconstructs the graph.

The call shapes define responsibilities, not a complete API implementation. Error codes, canonical serialization, ports and schema hashes must be fixed before interoperation. No process memory address is a persistent identity, and no illustrative reference in this example should be silently resolved to an arbitrary newest version.

35 Beam element appendix

A limited model preserved from the source with explicit assumptions

Use a straight prismatic two-node Euler-Bernoulli element of length L with DOFs [w1, theta1, w2, theta2], constant E, I, density rho_m and area A. The matrices below match the supplied Hermite formulation and were independently reconstructed by exact polynomial integration. They remain a model component requiring assembly and boundary constraints. [T L649-665; V2]

$$K_e = \frac{E I}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

$$K_{G,e} = \frac{P}{30L}\begin{bmatrix} 36 & 3L & -36 & 3L \\ 3L & 4L^2 & -3L & -L^2 \\ -36 & -3L & 36 & -3L \\ 3L & -L^2 & -3L & 4L^2 \end{bmatrix}$$

$$M_e = \frac{\rho_m A L}{420}\begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}$$

P is positive compression in this convention and the linearized tangent contribution is K_e - K_G,e. Mass and stiffness entries have mixed translational/rotational units because the DOFs have different units. Define scaling before comparing eigenvalue tolerances across meshes or coordinate conventions.

Implementation checks

Verify symmetry, rigid-body modes before constraints, positive mass on admissible modes, constrained static response, mesh convergence and a known buckling benchmark. Separate a model's instability onset from postbuckling, fracture and contact. Variable properties require suitable integration rather than substituting one unjustified average.

The exact checks of these element formulas do not validate the source's determinant rule or a physical specimen. The generalized eigenproblem in Section 22 uses its own constrained assembly and consistent sign convention. [X6-X7]

36 Primary references for core mechanics

Technical sources used to resolve consequential claims

References were consulted on 30 October 2026. They support the specific statements cited in the text; they do not validate the full proposed TOMOSS.A.D. implementation. Source titles below are followed by stable public URLs.

Concurrency

[X1] C++ working draft, atomic shared_ptr specialization. Supported operations and implementation-defined lock-free property. https://eel.is/c++draft/util.smartptr.atomic.shared

[X2] C++ working draft, memory ordering. Release/acquire synchronization conditions. https://eel.is/c++draft/atomics.order

[X3] C++ working draft, atomic smart-pointer general requirements. Ownership increments and deletion/deallocation sequencing. https://eel.is/c++draft/util.smartptr.atomic.general

[X4] Linux kernel documentation, What is RCU. Removal, reader completion and reclamation are separate phases. https://docs.kernel.org/RCU/whatisRCU.html

Rendering and structures

[X5] Pharr, Jakob and Humphreys, Physically Based Rendering, Microfacet Models. Primary author reference for the structure of microfacet reflection models. https://www.pbr-book.org/3ed-2018/Reflection_Models/Microfacet_Models

[X6] Applied Mechanics, Jönköping University, Buckling. Elastic column assumptions, eigenmodes and onset limits. https://mechanics.ju.se/SolidMechanics/Buckling.html

[X7] Bleyer, Numerical tours of continuum mechanics using FEniCS, Eulerian buckling of a beam. Implemented eigenproblem and constrained-degree treatment. Its demonstration uses a Timoshenko model; it supports the eigenproblem method, not the exact element choice in Section 35. https://comet-fenics.readthedocs.io/en/latest/demo/beam_buckling/beam_buckling.html

Energy and information

[X8] NASA Glenn Research Center, First Law - Internal Energy. Heat/work sign convention and energy accounting. https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/first-law-internal-energy/

[X9] MIT Unified Engineering, Thermodynamics and Propulsion, section 5.1. Isolated-system total energy and entropy. https://web.mit.edu/16.unified/www/FALL/thermodynamics/notes/node37.html

[X10] Shannon, A Mathematical Theory of Communication, 1948, author paper hosted by Harvard Mathematics. https://people.math.harvard.edu/~ctm/home/text/others/shannon/entropy/entropy.pdf

37 Primary references for deployment

Official platform sources and the implementation boundary

Device and host platform

[X11] POCO, X7 Pro specifications. Published processor, GPU, memory/storage options, connectivity and operating-system family. https://www.po.co/global/product/poco-x7-pro/specs/

[X12] Android Developers, Overview of background work. Select execution mechanisms for the actual application lifecycle. https://developer.android.com/develop/background-work/background-tasks

[X13] Android Developers, Dynamic code loading. Guidance for code provenance and avoiding unjustified dynamic loading. https://developer.android.com/privacy-and-security/risks/dynamic-code-loading

[X14] Android Developers, Permissions on Android. Explicit permission and access boundaries. https://developer.android.com/guide/topics/permissions/overview

[X15] Khronos, Vulkan Guide, Synchronization; and Android Open Source Project, Implement Vulkan. Explicit GPU synchronization and actual hardware/driver capability requirements. https://docs.vulkan.org/guide/latest/synchronization.html ; https://source.android.com/docs/core/graphics/implement-vulkan

[X16] Android Developers, Overview of memory management. Application memory limits and process lifecycle. https://developer.android.com/topic/performance/memory-overview

What this master establishes

The system has a defined numerical core, typed data and control contracts, per-instance state ownership, a runtime graph construction mechanism, a coherent publication model, concurrent job policies, recoverable commitment, optional domain modules and explicit acceptance gates. This is sufficient to begin implementation without guessing how the principal parts connect.

What remains a measured implementation result

Integrated runtime correctness, deployment compatibility, performance, local model capacity, sensor access, calibrated mechanics and physical predictions remain to be demonstrated on named versions, workloads and devices. The preserved numerical verification is a foundation for that work and is not substituted for it.

Future revisions should update the decision, schema and evidence records together. Preserve old profile meanings and artifact identities. Every newly claimed capability should link to the implementation version and acceptance evidence that established it.

38 Version 0.2 implementation map

Selected defaults that connect the architecture to a first build

PROPOSED. Sections 38-60 extend the retained master into implementation contracts. The initial target is a C++20 runtime inside an Android app, with a portable CPU reference build. This is a selected design profile, not a statement that the runtime, JIT or phone deployment already exists.

Decision	Version 0.2 baseline
State ownership	One ECS world per model instance; one owner of state-changing attempts; parallel child tasks use immutable inputs and private outputs.
Published unit	One coherent plan, root model state, world snapshot and generation. All reachable resources remain valid for every reader.
ECS implementation	Portable schema/query contracts first. A simple reference world is sufficient; Flecs or EnTT may be adapted after the same conformance gates.
Code execution	P0 packaged operators and descriptor compilation. P1 interpreted typed IR and P2 native JIT are separately gated extensions.
Record identity	Versioned canonical control records, exact legacy payloads, content digests and complete computation keys.
Durable authority	One local SQLite database and one writer coordinator. Immutable artifacts, accepted jobs, results, activation and outbox are recoverable.
External sources	Pinned model/data artifacts and event envelopes. Capability absence, stale data and unknown values remain explicit.
First vertical slice	The two-entity exact fixture in Section 56, including a replay, crash and publication test harness.

Compatibility boundary

tomoss_execution_v2 and tomoss_state_descriptor_v2 add world and system contracts under new schema identities. They do not reinterpret v1 records. The log net D transport frame, integer_bits_v1 and retained Q16 semantics remain unchanged. A v1 import either has an explicit mapping with provenance or is rejected as unsupported.

The runtime plan is a graph of executable modules and interfaces. An ECS world is the versioned data those systems operate on. A change to entity data does not automatically change the root sS or select a new LUT row: the descriptor declares that aggregation and transition. Sections 39-43 define this connection.

39 ECS identity and component schemas

Persistent entities and transient handles have different contracts

PROPOSED. One model instance owns an ECS world containing logical objects such as specimen elements, source channels or scene items. Its controller remains the sole state-changing authority. Separate runtime.module implementations, ecs.component data schemas and ecs.system operations; the word component must not conceal these different responsibilities.

Identity or schema	Required contract
EntityRef	Persistent world_id and canonical decimal entity_serial string. Allocate serials monotonically; compare serials numerically; never reuse a committed identity.
EntityHandle	Transient arena_instance_id, slot and generation. Reject another arena or stale generation; retire a slot before generation wrap.
ComponentSchema	Component ID, schema version/digest, fields, scalar profiles, units/frame requirements, validity and canonical encoding.
Component value	Validated value or immutable record reference. Missing component, invalid sample and valid zero remain distinguishable.
Relationship	Named relation schema and stable entity references; validate target existence and the declared topology rules.

Creation and replay

Workers emit provisional creation tokens from the logical transition, system, logical partition and local creation ordinal. The owner assigns serials in canonical token order at the merge barrier. The allocation counter and mapping commit with the world. Retries of a committed transition reuse its result; aborted candidates do not advance the committed counter.

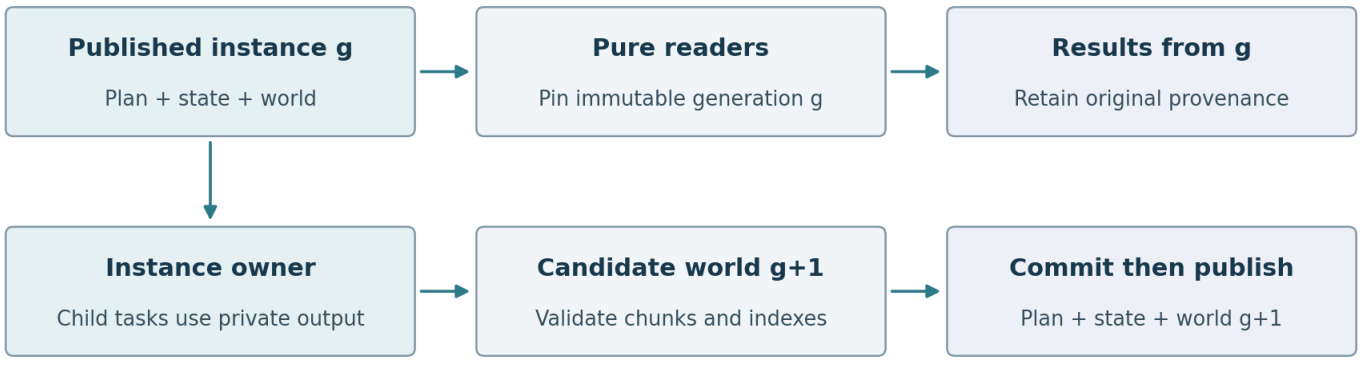
Retain the numerical profiles

A CoordinatePayloadRef points to the unchanged 28-byte coordinate frame and its envelope. PhaseState and Q16Signal retain their exact domains, validity and scale rules. A wider native integer container does not expand a 39-bit angle domain. Exact words never pass through floating-point JSON numbers or GPU display values merely to enter ECS storage.

Persist EntityRef, not a pointer or slot handle. After restart, rebuild transient mappings from the restored world. Generation-based recycling is documented by Flecs; the persistent/transient identity split above is a TOMOSS.A.D. design contract. [Y1]

40 Immutable worlds and borrowed views

Plan, state and world are published as one coherent generation



Proposed ownership model. Readers pin an immutable published world; child tasks prepare private outputs for the instance owner.

PublishedInstance pins plan_ref, state_ref, world_snapshot_ref and generation together. WorldSnapshot contains entity/revision indexes, schema manifest, immutable component chunks, relationships and allocation counter. Its logical digest excludes physical addresses and chunk layout. An archetype is a sorted set of component schema identities.

Borrow and mutation boundaries

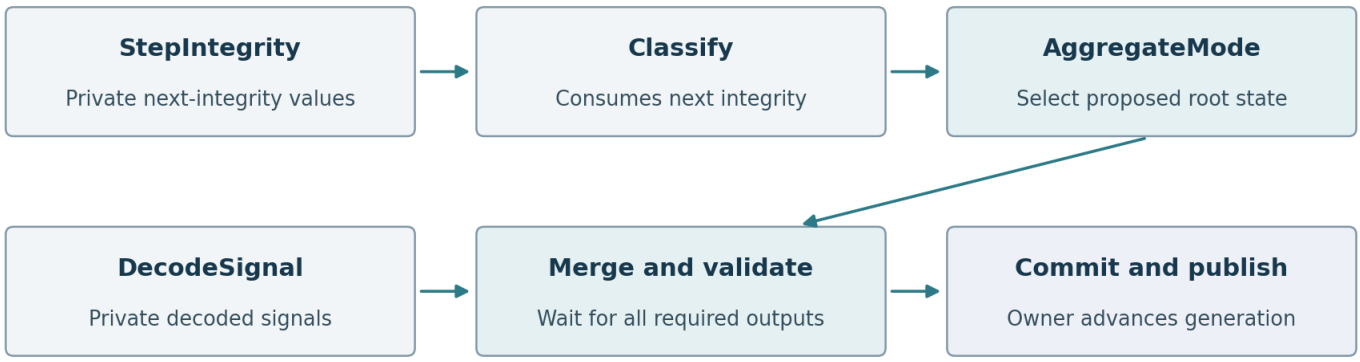
A BorrowedView carries its snapshot owner, query version, partition, access mode and borrow epoch. Raw references cannot escape its lifetime. Parallel tasks read pinned data and write private or proven-disjoint candidate buffers. Structural changes occur after candidate views are released at the declared barrier.

Old views remain valid while their immutable snapshot is retained. A newer deletion must not alter old membership or liveness data. Candidate construction uses explicit copying or copy-on-write with no mutable aliases. A const outer pointer does not freeze reachable mutable storage.

Flecs readonly execution can still write existing component values; its staging mechanism alone does not provide this snapshot contract. EnTT also documents view and synchronization constraints. Initialize required storage during plan construction and wrap library behavior behind these ownership rules. [Y1]

41 Systems and deterministic scheduling

Dependencies include buffer versions and entity regions



StepIntegrity and DecodeSignal can overlap; dependency edges remain fixed

Illustrative schedule. Independent integrity and signal tasks may overlap; classification waits for its producer, and the owner merges after all required outputs finish.

PROPOSED. A SystemContract names its query, implementation version, read/write sets, structural commands, resource effects, partitions, reductions and numerical policy. An access region includes world, component/resource, buffer version and entity range. Query membership, relationship traversal and shared globals also require declared access.

$$W_i \cap \left(R_j \cup W_j\right) \neq \varnothing \; or \; W_j \cap R_i \neq \varnothing$$

For overlapping regions, this hazard requires a declared dependency or plan rejection. Unknown disjointness is treated as conflict. Reading a base buffer while writing a private next buffer is separate storage; a consumer of the next value still needs its producer edge.

Stable work with flexible workers

Freeze matching entities in EntityRef order. Logical partitions and reduction slots are independent of physical worker count. Use a fixed reduction tree, checked arithmetic and named empty-partition behavior. Exact core rules remain authoritative; approximate backends need their own conformance profile. Child tasks prepare outputs; only the instance owner commits and publishes.

42 Structural commands and merge barriers

Membership changes are validated before they become visible

PROPOSED. Every bounded command buffer identifies its base world generation, logical transition, system and partition. A command is an intention to construct the next world; it does not modify the published world or become visible midway through a query.

Command	Validation at the owner barrier
CreateEntity	Resolve a provisional creation token and validate initial component schemas and references.
DestroyEntity	Check persistent identity and expected entity revision. Reject conflicting writes to the same deleted entity.
AddComponent	Require absence and a valid value under the exact declared schema.
SetComponent	Require presence, compatible schema and expected revision; preserve quantity/profile meaning.
RemoveComponent	Require presence under the initial strict policy; removal is distinct from storing an invalid value.
SetRelationship	Check relation schema, target identity, cardinality and any cycle restrictions.

Merge protocol

Wait for required tasks and release candidate borrows. Collect commands by the plan's canonical system/partition/local order. Validate access ownership, domains, capabilities and resource limits. Reject overlapping intents unless the descriptor names a merge operator; last-writer-wins is never an implicit fallback.

Apply the accepted commands to a private candidate world, rebuild affected query indexes and check its invariants. Commit the world, next model state, result, activation and outbox together. Publish the coherent instance root only after durable commit. Failed construction or migration preserves the previous published generation.

Lifetime and module boundaries

Retain old chunks, schema definitions and implementation modules while readers or device submissions use them. Export stable IDs and typed buffers across plugin boundaries; keep ECS-library pools and allocator ownership internal. These are project ownership requirements for any selected ECS library.

A command-buffer limit produces resource_limit before publication. Replay deduplicates the logical transition; it does not turn malformed commands into successful operations.

43 Descriptor v2 and typed intermediate form

Explicit successor schemas extend the original control contract

PROPOSED. tomoss_state_descriptor_v2 adds ECS world, query and scheduling contracts. tomoss_execution_v2 binds the admitted job to world_snapshot_ref and the corresponding schema/system manifests. Version 1 remains a separate supported contract; no existing descriptor or execution record acquires v2 meaning silently.

Descriptor v2 field group	Required information
Registries and world	Exact component-schema, system and runtime-module registry references; world schema and compatible backend.
Queries and bindings	Required/excluded components, relationship scope, input/output ports, units and numerical profiles.
Access and schedule	Read/write/effect regions, producer edges, logical partitions, reduction rules and merge barriers.
Transitions and limits	Allowed target descriptors, explicit state/world migration, capability scope and complete resource budgets.

Compiler stages

```
DescriptorV2
  -> ParsedIR
  -> TypedGraphIR
  -> ScheduledPlanIR
  -> FrozenRuntimePlan
```

ParsedIR preserves source spans and rejects unknown fields. TypedGraphIR resolves exact schemas and implementations, validates ports, domains and units, and makes every conversion an explicit node. ScheduledPlanIR validates hazards, barriers, budgets and deterministic ordering. FrozenRuntimePlan retains the complete artifacts required to execute and inspect that graph.

The plan is complete before READY admission: no unresolved newest-version reference, hidden pool initialization or unvalidated kernel compilation remains on the worker path. Admit freezes the execution envelope, input manifest, state generation and world snapshot. PublishedInstance keeps plan, state and world coherent across publication.

A v1-to-v2 migration is a named operation producing a new descriptor and validation report. The retained coordinate, Q16 and integer/rational payload profiles are unchanged. The inspection diagram reads FrozenRuntimePlan and reports the exact query/system graph selected for that instance.

44 Operator ABI and resource ownership

Proposed native interface with explicit lifetimes

Every native operator shall expose a versioned C-compatible boundary. The registry binds that interface to an immutable implementation, input/output schemas, numerical policy and effect class. C++ classes, STL containers, compiler-specific exceptions and process addresses shall not become persistent or cross-module identities. A module is admissible only when its declared ABI matches the host.

Boundary field	Required behavior
Version and identity	ABI version, structure byte sizes, operator version and schema digests; reject an unknown required field or incompatible layout.
Input and output buffers	Pointer plus byte length, element schema, alignment and access mode. Validate bounds; decode canonical bytes explicitly.
Opaque handles	Host-issued identity with generation and owner. Reject stale handles; never serialize their process representation.
Allocation and errors	Host allocator supplies output storage. Return a stable status and host-owned bounded diagnostic; release through the same allocator.
Cancellation and effects	Poll a host cancellation token at declared safe points. Prepare effects as intents; the operator cannot commit authoritative state.
Module and device ownership	Retain module, snapshot and buffers until their last CPU caller, callback and GPU submission have completed.

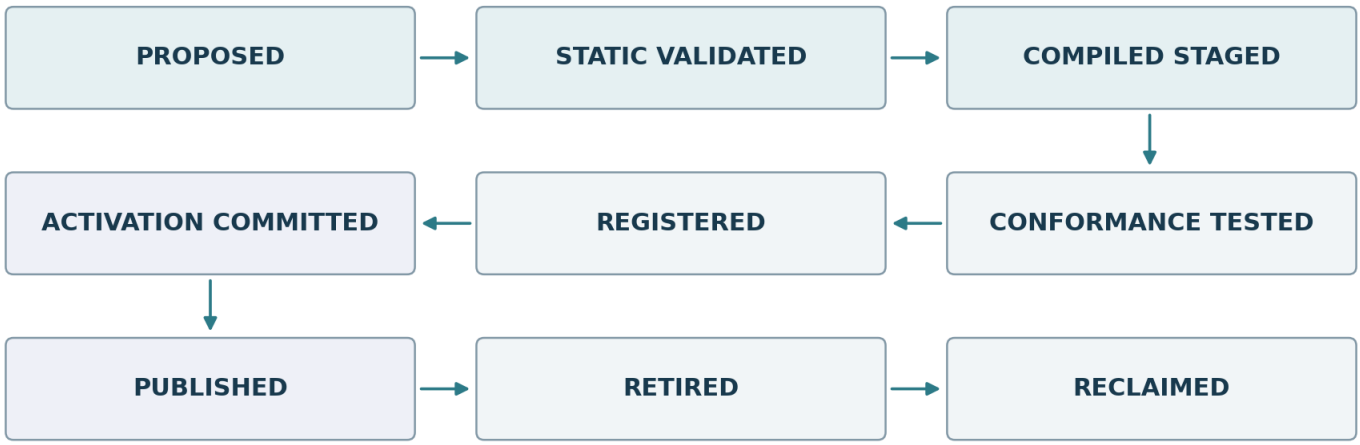
Call and completion are different events

An asynchronous call returns an owned completion handle. A successful submission means accepted work; it does not mean completed output. The host releases resources after observing completion and any required device-to-host visibility operations. Failed or cancelled work releases temporary resources and records a typed outcome. Cancellation has no power to erase an already committed result.

JNI shall remain a narrow controller boundary: JNIEnv is thread-local, and native workers shall not retain local Java references across their valid lifetime. Android ABI representation and the canonical TOMOSS.A.D. encoding remain separate. Vulkan completion and synchronization remain explicit even when memory is shared. These interface requirements are proposed; no ABI implementation has been validated here. [Y6; Y11]

45 Compiler tiers and module lifecycle

Proposed admission separates compilation from activation



Proposed compiler lifecycle. Registration permits validated plans to reference a module; durable activation and publication remain separate. Reclamation waits for retained users.

Tier	Scope
P0 baseline	Packaged AOT kernels; compile descriptors into typed plans.
P1 optional	Interpret a bounded typed IR with explicit resource limits.
P2 experimental	Local LLVM ORC native JIT after Android deployment gates.

ORC supplies JIT infrastructure, not a C++ frontend. Freeze frontend/toolchain, target triple, CPU features, ABI, DataLayout, dependencies and optimization policy. Validate IR, link privately, run conformance checks, then register an immutable artifact. A verifier is neither a sandbox nor semantic proof. Arbitrary native proposals require a separate trust/isolation decision. [Y5]

The exact profile shall introduce no unproved nsw/nuw assumptions or fast-math transformations. LLVM overflow flags can produce poison; floating-point flags change permitted semantics. Every failed stage preserves the active plan. [Y5]

Pin each module with its executing snapshot. ORC resource removal occurs only after all callers and dependent work finish; compilation or symbol lookup never authorizes publication. These are proposed gates. [Y5]

46 Model and dataset admission registry

Proposed import records preserve origin and executable meaning

The importer shall progress through DISCOVER, PIN, FETCH_TO_STAGING, CLASSIFY, VALIDATE, COMPATIBILITY_TEST and ADMIT. Each failure preserves its report and prevents admission. Repository type identifies a hosting category; artifact kind identifies actual content. A repository labelled model can contain documents, and a dataset repository need not contain training rows.

Manifest group	Fields to freeze before admission
Origin and bytes	Repository type/ID, resolved full commit, file paths, byte sizes and cryptographic digests.
Model interpretation	Architecture/configuration, tensor schema, format/version, dtype or quantization, admitted runtime and backend.
Text interpretation	Tokenizer files, vocabulary/merges, special tokens, chat template and preprocessing digests.
Dataset interpretation	Configuration, splits, schema/features, formats, row identity/order policy and transformation versions.
Use and resources	License declaration or license-file digest, review status, context limits, memory budget and requested capabilities.
Evidence	Validation report, test-set digest, provenance and compatibility result; distinguish declarations from measured checks.

Hugging Face supports branch, tag and commit revisions. Admission shall resolve mutable names to a full commit and verify the staged files. Cards supply descriptive metadata; tensor containers such as safetensors still require a compatible architecture and runtime. Downloading a repository shall not execute its scripts or grant application capabilities. [Y7]

Concrete discovery record

The public dataset card c0mbatduckzz/TOM-GROK describes a historical research-artifact archive and points to a model-format publication. Record repo_type=dataset and artifact_kind=research_archive as a publisher declaration. Its status here is DISCOVERED_ONLY: no full revision, file digests or executable compatibility have been established. This example contains no downloaded or verified executable and is not an admitted model. [Y8]

Offline execution shall use an already verified snapshot. Missing bytes, an unresolved license status or an incompatible backend returns a typed admission failure instead of silently selecting another revision.

47 Finite context and memory budgets

Proposed admission uses measured resources and exact tokenization

Every generation request shall reserve its complete formatted prompt and output allowance before execution. Count tokens with the admitted tokenizer and template; count formatting once. The following equations define proposed admission checks, not a performance measurement or a verified device capacity.

$$T_{sys}+T_{history}+T_{retrieved}+T_{input}+T_{output}+T_{format}\leq C_{eff}$$

$$C_{eff}=min\left(C_{model},C_{backend},C_{policy},C_{memory}\right)$$

The model limit must be supported by its architecture/configuration and validation. A tokenizer can expose a very large default sentinel when its limit is unspecified; that value shall not establish usable capacity. Reject, explicitly truncate permitted fields, summarize with provenance, or reduce retrieval when a request exceeds its allowance. [Y13]

Conditional KV-cache estimate

$$B_{KV}\approx 2\,L\,T\,H_{KV}\,d_h\,b\,N_{batch}$$

For conventional dense autoregressive attention, L is layers, T cached tokens, H_KV key/value heads, d_h head dimension, b bytes per stored value, and N_batch batch size; two accounts for keys and values. Sliding windows, grouped heads, quantization, padding and cache implementations change the estimate. Measure their actual allocation and include workspaces separately. [Y13]

$$B_{weights}+B_{KV}+B_{work}+B_{runtime}+B_{GPU}+B_{stage}+B_{margin}\leq B_{admitted}$$

Count each underlying allocation once when CPU and GPU share storage. Include compilation, conversion and loading peaks. The phone's marketed 12 GB is not the process budget; B_admitted is a measured application allowance with a margin. Lower batch size or reject admission before exceeding it. Admission shall also bound concurrent jobs and retrieval bytes. A per-request limit alone does not constrain total memory when multiple instances run, and a cached result still needs identity validation.

An inference key includes model, tokenizer/template, inputs, retrieval, generation settings/seed and backend profile. A seed alone does not prove deterministic execution: memoize as pure only under an established reproducibility contract; otherwise label reuse as a stored prior response. External memory extends accessible information, not active context. Eviction preserves authoritative sources/results.

48 Canonical records and computation identity

Exact values survive serialization and retain their meaning

PROPOSED. Adopt tomoss_c14n_jcs_v1 for new control and execution records. Bound parsing depth/size and validate schemas before canonicalization. Retained log net D payloads keep their original bytes and profile; the envelope references their raw-byte digest.

Representation	Required canonical rule
Fixed-width words	Preserve declared signedness, width and leading zeros in binary strings. Never route wider values through JSON floating numbers.
New rational values	Use a positive denominator, reduced numerator/denominator and canonical zero/sign under the named rational schema.
Objects and strings	Reject duplicate keys, invalid Unicode and nonfinite numbers. Apply JCS property ordering and UTF-8 output; preserve string data.
Arrays and optional data	Preserve array order. Canonicalize a declared set explicitly. Missing, null, invalid and zero remain distinct schema cases.
Raw and approximate data	Preserve raw-source identity separately. Approximate values need a named format/rounding policy or explicitly tagged floating-point bits.

Domain-separated content identity

```
D = SHA256(UTF8("TOMOSS:JCS:1") || 0x00 ||
           UTF8(kind) || 0x00 || JCS(record))
```

Constrain kind to the declared ASCII identifier alphabet without NUL; include schema ID/version in the record. RFC 8785 defines serialization, not record meaning. Its number path uses IEEE 754 double precision: wide exact values require validated strings or byte references. Schemas may explicitly permit bounded JSON integers in the exact safe-integer domain. [Y2]

Identity has several purposes

An intent ID identifies a request; an attempt ID identifies one execution; a content digest identifies bytes or canonical meaning. A pure computation key additionally binds operator/implementation, inputs, parameters, model, calibration, numerical policy and relevant backend profile. Acquisition time and coordinate metadata stay included when they affect the result.

Completion timestamps belong outside the canonical numerical-result projection. Cache reuse still preserves each intent and its provenance. A digest does not authenticate a producer, grant a capability or turn matching command payloads into the same intended action.

49 SQLite persistence profile and schema

A proposed local transaction boundary for authoritative state

The initial profile uses one runtime process and one writer coordinator. Keep authoritative state, results, activation records and outbox entries in one local SQLite database. This is a backend proposal; no integrated runtime or crash benchmark is claimed.

Tables	Keys and owned information
artifact and snapshot	Immutable digest-keyed bytes; schema/kind/length; snapshot binds instance, generation, plan, world/state record and registry epoch.
request_intent and job_attempt	Unique intent and attempt IDs; request digest; frozen input manifest; mode, admission sequence, status and cancellation request.
instance_head and activation	Current snapshot/generation per instance; unique instance-generation activation linked to its originating attempt or control intent.
job_result and journal	One authoritative result per attempt; unique event IDs and monotonic journal sequence with canonical event bodies.
outbox and delivery_attempt	Stable event/destination/body identities; exact result/generation; delivery attempts, acknowledgements and retry decisions.
sequence_budget	Checked counters, persisted admission reservations and ordered-commit cursor. Exhaustion fails without wrapping.

Database configuration is an observed contract

Target WAL with synchronous=FULL; enable foreign_keys on every relevant connection and verify settings. Bound busy handling. Require a tested build with the WAL-reset fix: 3.51.3 or later, or backports 3.44.6/3.50.7. Record linked version/source ID and compile options. Durability depends on the storage/VFS honoring synchronization; the phone's system build is not assumed. [Y3-Y4]

Storage and computation lifetimes

Store bounded authoritative metadata/state records as BLOBs initially. Wide exact values remain strings/BLOBs; checked SQL integer counters do not change numerical payload profiles. Large external artifacts require durable publication before a transaction references them, plus retention while jobs or checkpoints depend on them.

Workers retain immutable application records, not long SQLite read transactions or open cursors throughout computation. Bound checkpoint/WAL growth. WAL transactions across multiple attached databases are not one atomic unit; keep the authoritative commit together. Foreign keys enforce references, while application validators enforce their semantic meaning. [Y3-Y4]

50 Admission, commitment and publication

The durable journal and the active snapshot advance coherently

Each instance owner admits at most one unresolved mutation. A successor binds input after its predecessor terminates and any committed activation is published. Instance-affecting configuration changes use the same owner. Pure snapshot jobs may execute concurrently.

Boundary	Proposed required action
Freeze and reserve	Resolve dependencies; pin one complete plan/world snapshot and input manifest; validate capabilities; reserve bounded admission capacity.
Admit durably	BEGIN IMMEDIATE; resolve intent idempotency. Under the owner, a mutation requires pinned base = session-published generation = durable head and claims the unresolved-mutation slot. Persist READY, sequence, reservation and journal; COMMIT.
Wake and execute	Acknowledge durable acceptance; notify Q0 by attempt ID. Execute into private working state and prepare validated output outside the write transaction.
Commit outcome	Atomically persist result, terminal attempt status/journal and required outbox after cancellation checks. Only mutations also write next state/world/snapshot and activation, updating exactly one expected-generation head row.
Publish activation	After successful commit, atomically publish the prebuilt coherent snapshot. Advance the session-local publication generation before admitting another mutation.
Deliver and retire	Dispatch eligible committed events. Retain old records, modules and device resources until their documented users finish.

The publication gap has an explicit gate

The mutation admission gate remains closed between activation commit and pointer publication. A persisted published boolean cannot prove that a restarted process has reconstructed its pointer. Recovery rebuilds committed heads before reopening their gates. Activation notifications wait until the current session has published at least their generation; every event identifies its exact immutable result/generation.

Old snapshots remain valid inputs

A stale frozen mutation requires explicit rebinding and validation before admission. Pure jobs may retain older snapshots at admission and commit. The durable READY registry is authoritative and Q0 notification is idempotent: a crash or missed wakeup cannot lose accepted work.

BEGIN IMMEDIATE can return busy; retry within budget. Keep compilation, GPU waits, network calls and observer callbacks outside write transactions. Treat a busy COMMIT according to the connection state, rather than blindly rerunning the entire transaction. [Y3]

51 Cancellation, recovery and delivery

One authoritative outcome survives interruption and retries

Cancellation and commitment are serialized decisions. A cancellation recorded before the commit check can win; a later request reports too_late and cannot erase a committed result. Failed/cancelled accepted attempts still receive terminal sequence records so ordered commitment can advance.

Interruption or retry	Recovery decision
Before admission commit	No accepted execution attempt exists. Retry the same intent without creating an uncontrolled duplicate.
After READY before Q0 wakeup	Read durable READY attempts and re-notify using their existing IDs.
During execution	Reconstruct frozen dependencies; rerun pure work or resume a compatible checkpoint under its original profile.
During result transaction	Recover the complete result/state/activation/outbox set or none of it. Never expose a partial authoritative outcome.
After activation before publication	Rebuild the committed snapshot, publish it and reopen the instance gate; old-session publication observations are insufficient.
After recipient action before acknowledgement	Retry the same event ID when the recipient deduplicates; otherwise reconcile the uncertain effect.
Lost or ambiguous COMMIT response	Inspect transaction/connection state and query the attempt/result identity before rerunning. A lost response does not prove rollback.

Duplicate requests and external effects

The same intent ID and request digest return existing status; a different digest under that ID fails with idempotency_conflict. A new intent may represent a new action despite equal payloads. External action IDs remain stable across delivery retries. Local atomic commitment does not establish exactly-once physical action.

Recovery retains its evidence

Storage exhaustion pauses/faults accepted work recoverably; do not claim a terminal record that was never written. Missing original artifacts fail explicitly rather than selecting newer versions. Use a consistent SQLite backup plus all externally referenced artifacts; copying only a live database file is insufficient for a WAL backup. [Y4]

Acceptance injects crashes at every boundary above, races cancellation with commit, restores old snapshots and repeats delivery after a lost acknowledgement. These are required tests of the proposed backend, not results already demonstrated.

52 Telemetry and event-time contracts

Capture observations without confusing arrival with physical time

PROPOSED. A source adapter emits immutable observations before a model consumes them. A phone sensor, camera, network topic and archived dataset each have their own clock and acquisition contract. A live connection does not establish that a received measurement is recent, calibrated or complete.

Envelope field	Required interpretation
Source identity	Adapter/version, channel, source session or boot identity, source sequence and stable event ID where supplied.
Acquisition time	Original clock domain, exact timestamp, interval/exposure where applicable, clock-mapping reference and mapping uncertainty.
Receipt time	Local monotonic receipt time and durable arrival sequence; retain UTC only with a named time mapping.
Values and quality	Typed payload; units and coordinate frame; calibration; validity, saturation, missingness and quality flags.
Stream control	Window policy, lateness bound, watermark rule, deduplication rule, gap policy and bounded buffer/reservation.
Replay identity	Payload digest, selected source interval and all transformations. Replaying a stream pins these records rather than refetching a moving source.

Clock and ordering rules

Android SensorEvent timestamps use nanoseconds and the elapsedRealtimeNanos time base. They are not UTC timestamps. The adapter records a boot/session identity and any mapping into a shared time frame. Camera time-base metadata must be checked separately before camera and motion events are aligned. [Y12]

The descriptor chooses processing-time or event-time windows, explicit late-data treatment and a stable tie-breaker. A provisional window and a finalized window are different record versions. Late correction creates a derived result; it does not silently rewrite a committed history. Missing sequences remain observable.

MQTT adapter

MQTT is an optional payload transport. Its QoS levels govern protocol delivery; application effects still need the stable outbox event ID and recipient deduplication/reconciliation. A broker acknowledgement does not prove a model committed or a device acted. Bound incoming queues and record overflow gaps under the selected capture policy. [Y9; proposed application contract]

53 Topology and geospatial ingestion

Keep mathematical structure, terrain and execution graphs distinct

Meaning of topology	TOMOSS.A.D. representation
Runtime topology	Typed modules, ports and dependencies in an immutable runtime plan. Derived inspection view; Section 11.
ECS relationships	Versioned relations among EntityRefs, with ownership and cycle policies where needed.
Geospatial structure	Coordinate reference, terrain/surface samples, masks, mesh adjacency and spatial indexing.
Algebraic structure	A specified set, operation and invariant. The retained D6 action is one explicit automorphism contract; Section 7.

Concrete AHN3 source profile

Earth Engine exposes AHN/AHN3 as an ImageCollection with dtm and dsm elevation bands in metres at 0.5 m pixel size. Its catalog describes aircraft/helicopter laser acquisition and a 2014-2019 coverage period. DTM represents ground; DSM represents the highest surfaces. It is historical elevation data. A live satellite or phone feed requires a separate adapter and acquisition record. [Y10]

Proposed terrain pipeline

Select an area and source images; freeze image IDs and export request; record collection, band, acquisition metadata and processing date. Query the actual projection, transform and vertical reference. Record axis order, units, pixel-centre convention, nodata mask and the exact resampling operation. Missing metadata blocks transformations that require it.

Export bounded tiles, preserve their bytes and digests, then validate dimensions and seam overlap. Derive a mesh or sampled height field under a named algorithm and level-of-detail policy. Cache by source bytes, coordinate/vertical transforms, resampling and mesh profile. A display tile is not a new calibrated measurement.

Do not insert AHN heights into an ellipsoidal coordinate formula until their height reference has been established and the required transformation is available. Derive slope or normals with explicit spacing, boundary treatment and uncertainty. Phone location and terrain elevation remain separate observations until their frames and time scopes are reconciled.

A geospatial tile may be an ECS entity referencing immutable arrays; the arrays need not become millions of individually scheduled entities. Earth Engine access, export quotas and network availability are deployment capabilities, not assumptions embedded in the numerical core. [Proposed adapter design]

54 GPU rendering and Bayer processing

Two image paths with explicit buffers, metadata and completion

Committed world to display

PROPOSED. A view job pins a committed world, camera/view transform and display profile. It produces bounded visible tiles or draw batches, performs the chosen shading/compositing steps, and records the originating generation. The UI may display a newer completed frame while an older immutable result remains in the journal.

A Vulkan backend queries and enables supported features, declares attachment formats and load/store operations, and schedules resource transitions and completion. Tile dimensions are a measured backend choice; the application does not assume control of a Mali implementation's internal tiles. Device resources and referenced modules survive until their relevant submissions complete. [Y11]

Raw camera to linear image

Android RAW_SENSOR needs camera metadata. Admit Bayer demosaicing only for a supported CFA/capture mode; other arrangements need their own processing. Read layout, crop origin and applicable per-frame black/white levels. YUV/JPEG outputs already have different processing semantics. [Y12]

Processing stage	Contract and acceptance evidence
Unpack and normalize	Respect row/pixel layout, sample range, crop and black/white levels. Flag clipping; retain raw bytes and capture metadata.
Demosaic	Named interpolation method, CFA phase, border rule and rounding profile. Use a documented halo for tiled neighborhoods.
Colour and view	Named gain, colour transform, working space and tone/display conversion; record which processing the camera already applied.
Publish output	Confirm GPU completion and required host visibility before committing readback. A screen-only derived view retains generation/provenance separately.

A simple bilinear CPU reference may establish interpolation fixtures; an accelerated method must match its selected exact or tolerance profile. Test constant mosaics, colour-channel placement, odd crop origins, edge halos, strides and saturated values. Tiled and whole-image paths must agree under that profile. Demosaicing estimates missing colour samples; it does not recover independently measured RGB values at every pixel.

GPU failure produces a typed outcome. A CPU fallback records the new backend/profile and its evidence; it never silently reuses an incompatible cache key. Numerical simulations and camera image processing remain separate systems even when both produce textures. [Proposed conformance rules]

55 Android app and native deployment

Proposed POCO target profile requires measurements on the device

The POCO X7 Pro 12 GB remains a candidate deployment target. Its installed OS, process ABI, available memory, driver features and sustained performance have not been observed here. Native C++ operates inside Android's application sandbox; Android retains scheduling, permissions, memory mapping and device access. “Bare metal inside an app” therefore names an implementation style, not privileged hardware ownership. [Y6]

Deployment gate	Evidence and failure behavior
Packaged baseline	NDK-built libraries match the process ABI; JNI versions and lifetimes pass tests. An incompatible build does not load.
Optional native JIT	Verify target/runtime support, relocations, executable-memory behavior and instruction-cache handling. If unproven, disable P2.
Distribution and imports	Review the actual dynamic-code design for its distribution channel; verify origin and integrity before admitted loading.
Vulkan backend	Query API version, features/extensions, queue families, limits, formats and memory types; enable only supported required features.
Memory and lifecycle	Measure allocation peaks; test cancellation, backgrounding, process death and recovery. Restore committed records.
Sustained workload	Record workload, OS/build, backend, latency, memory and thermal behavior. Select budgets from measurements.

The Android controller shall own lifecycle and platform services; a narrow JNI bridge shall connect it to the native runtime. Packaged operators are P0. P1 and P2 remain separately admitted options. Android warns that many forms of dynamic loading, especially from remote sources, can violate distribution policy; that is a concrete gate, not a claim that ORC already works on this phone. [Y6; X13]

GPU discovery precedes backend creation. Record the enabled feature set and numerical profile with results. A CPU pointer publication does not complete GPU work; retain buffers and modules through submission completion and required visibility operations. Fallback must preserve the requested semantics or return unsupported, and its selected backend must appear in provenance. These deployment requirements remain unimplemented acceptance gates. [Y11]

56 End-to-end ECS conformance fixture

Expected exact results for a two-entity transition

Profile ecs_integrity_demo_v1 defines this fixture and its STABLE, BOUNDARY and ONSET codes. This section specifies expected outputs; Sections 69-72 report a bounded serial reference evaluation. Store integrity in integer units of 1/20; subtract one unit per step, clamp at zero and compare with threshold 3. Use positive s0 = 1 signal unit for the retained Q16 endpoint mapping.

Entity	Integrity units	Next class	Q16 code to signal
E41	4 -> 3	BOUNDARY	0 -> 0
E42	3 -> 2	ONSET	65,535 -> 15

$$X_{n+1} = max\left(0, X_n - 1\right), trigger_{n+1} = \left[X_{n+1} < 3\right]$$

Schedule and publication

StepIntegrity and DecodeSignal run independently into private output slots. Classify consumes the new integrity values; AggregateMode selects ONSET if any valid entity is ONSET, otherwise BOUNDARY if any is BOUNDARY, otherwise STABLE. Missing or invalid required inputs produce an unavailable result, not a normal-state default.

The owner merges deferred classification-tag changes after the barrier and validates the candidate world. Result, next state, world, activation and outbox commit together. New admission sees the new root after publication. A diagnostic already pinned to the old world still reads integrity values 4 and 3 and may commit that labeled result later.

Required edge checks

Repeat with one and several physical workers and varied completion order. Canonical exact outputs must agree. Reject an unordered second writer to the same next-integrity region, a stale handle, missing target schema and nonpositive s0. Exhausting the command buffer must preserve the old root. Deleting E41 in a later generation must not invalidate its retained old-world view.

Compare decoded endpoints using exact words/rationals. Structural tags remain categorical data. Persisted identity and canonical state must remain unchanged by chunk compaction or a different physical pool layout. These are acceptance expectations for the proposed ECS adapter.

57 Work packages and release gates

A concrete build order with observable completion criteria

Work package	Deliverable and exit gate
WP1 Records and exact core	Immutable payload/envelope codecs, strict semantic validation, canonical digests and version mappings. Retain the Section 29 baseline; add wide-integer round trips and malformed-input fixtures.
WP2 Reference world	Persistent entities, component registry, immutable snapshots and private candidates. Reject stale handles and prove old-reader isolation across deletion/migration.
WP3 Plans and execution	Descriptor v2, typed dependencies, bounded Q0, reference scheduler and deterministic command merge. Execute the Section 56 oracle and reject conflicting writes.
WP4 Durable coordination	Selected SQLite build, intent/attempt/result schema, admission/commit/publication gates and outbox. Inject crashes at every boundary in Section 51.
WP5 Android vertical slice	Lifecycle controller, narrow native bridge, capability report, committed-result view and recovery after process death. Measure peak memory and sustained behaviour on the actual phone.
WP6 Optional extensions	One pinned model or dataset, one source adapter and one renderer profile at a time. Admit each only with compatibility evidence; stage P1/P2 compiler tiers separately.

Release evidence package

A conforming release includes source/build identities; schema/profile manifest; descriptor and graph exports; numerical and ECS oracle outputs; race/crash results; dependency and capability report; retention/recovery procedure; and a named-device performance report. Record test failures and unsupported capabilities alongside successes.

Replay tests vary worker counts and interleavings. Exact or certified bitwise profiles reproduce canonical values; approximate profiles meet their named tolerances. The selected mode fixes commitment ordering. Concurrency tests directly observe aliasing, structural barriers, module retirement and the gap between durable activation and visible publication.

Status at this revision

Section 29 retains the numerical verification; Section 72 adds the small serial reference evaluation. ECS integration, compiler, persistence, telemetry, GPU and deployment still require their stated acceptance evidence. No integrated runtime or phone benchmark is claimed. OOB and AC remain unresolved labels; later definitions must not silently change an existing schema.

58 References: ECS and persistence

Primary documentation consulted on 30 October 2026

The Y-series supports the implementation extension. References describe external mechanisms; TOMOSS.A.D. ownership, publication and acceptance rules remain the proposed project contracts. A grouped key points to the relevant document within that group.

[Y1] ECS behaviour and library boundaries

Flecs, Entities and Components: entity lifetime and component operations. https://www.flecs.dev/flecs/EntitiesComponents.html

Flecs, Systems: queries, staging, scheduling and read/write behaviour. https://www.flecs.dev/flecs/Systems.html

EnTT, Crash Course: entity-component system: storage, views, iteration, snapshots and threading responsibilities. https://skypjack.github.io/entt/md_docs_2md_2entity.html

[Y2] Canonical control records

Rundgren, Jordan and Erdtman, RFC 8785, JSON Canonicalization Scheme (JCS), Informational RFC. Deterministic encoding and IEEE 754 number constraints. https://www.rfc-editor.org/rfc/rfc8785

[Y3] SQLite transactions and referential integrity

SQLite, Transaction: writer concurrency, BEGIN IMMEDIATE and transaction outcomes. https://www.sqlite.org/lang_transaction.html

SQLite, Foreign Key Support: connection configuration and referential constraints. https://www.sqlite.org/foreignkeys.html

[Y4] SQLite durability and recovery

SQLite, Write-Ahead Logging: reader/checkpoint interaction, one-database atomicity boundary and WAL-reset fix versions. https://www.sqlite.org/wal.html

SQLite, PRAGMA synchronous: WAL durability modes and synchronization requirements. https://www.sqlite.org/pragma.html#pragma_synchronous

SQLite, Online Backup API: consistent backup of a live database. https://www.sqlite.org/backup.html

The selected database build and configuration must be captured by the implementation. These citations establish a design basis; they do not certify an untested phone filesystem or storage stack.

59 References: compiler and AI artifacts

Toolchain, provenance and bounded model execution

[Y5] LLVM and the native compilation path

LLVM, ORC Design and Implementation: execution sessions, linking, modules and resource management. https://llvm.org/docs/ORCv2.html

Clang, Assembling a Complete Toolchain: frontend, runtime and link dependencies. https://clang.llvm.org/docs/Toolchain.html

LLVM Language Reference: verification, arithmetic flags, poison values and floating-point semantics. https://llvm.org/docs/LangRef.html

[Y7] Hugging Face artifact admission

Hugging Face Hub, Download files: repository type and revision selection. https://huggingface.co/docs/huggingface_hub/guides/download

Hugging Face, Model Cards: intended uses, limitations and metadata. https://huggingface.co/docs/hub/model-cards

Hugging Face, Dataset Cards: dataset documentation and metadata. https://huggingface.co/docs/hub/datasets-cards

Hugging Face, Safetensors: tensor-file format and library scope. https://huggingface.co/docs/safetensors/index

[Y8] Project-associated public discovery example

c0mbatduckzz/TOM-GROK dataset card, describing a historical research-artifact archive. Publisher statements inform classification; executable contents and compatibility were not established. https://huggingface.co/datasets/c0mbatduckzz/TOM-GROK

[Y13] Context and inference memory

Hugging Face Transformers, Tokenizer: tokenization and model maximum-length configuration, including sentinel defaults. https://huggingface.co/docs/transformers/main_classes/tokenizer

Hugging Face Transformers, Cache strategies: KV-cache tradeoffs and architecture-specific limits. https://huggingface.co/docs/transformers/kv_cache

The registry does not turn a repository category, card statement or filename into proof of runnable weights. Admission requires the concrete manifest, full revision, bytes and compatibility evidence described in Section 46.

60 References: mobile and source adapters

Host boundaries, data acquisition and graphics contracts

[Y6] Android native application boundary

Android Developers, Get started with the NDK. https://developer.android.com/ndk/guides

Android Developers, JNI tips. https://developer.android.com/ndk/guides/jni-tips

Android Open Source Project, Application Sandbox. https://source.android.com/docs/security/app-sandbox

Android Developers, Android ABIs. https://developer.android.com/ndk/guides/abis

[Y9] Messaging and [Y10] elevation data

OASIS, MQTT Version 5.0, 7 March 2019: publication, delivery and session semantics. https://docs.oasis-open.org/mqtt/mqtt/v5.0/os/mqtt-v5.0-os.html

Google Earth Engine Data Catalog, AHN3: Netherlands AHN 0.5m: collection, bands, acquisition period and units. https://developers.google.com/earth-engine/datasets/catalog/AHN_AHN3

[Y11] Graphics capabilities and synchronization

Khronos, Vulkan Guide, Enabling Features. https://docs.vulkan.org/guide/latest/enabling_features.html

Khronos, Vulkan Guide, Synchronization. https://docs.vulkan.org/guide/latest/synchronization.html

Khronos, Vulkan Specification, Render Pass: attachments and rendering configuration. https://docs.vulkan.org/spec/latest/chapters/renderpass.html

[Y12] Sensor time and raw camera metadata

Android Developers, SensorEvent.timestamp. https://developer.android.com/reference/android/hardware/SensorEvent#timestamp

Android Developers, CameraCharacteristics: CFA arrangement, black/white levels and timestamp source. https://developer.android.com/reference/android/hardware/camera2/CameraCharacteristics

Android Developers, ImageFormat.RAW_SENSOR: sample format and metadata dependency. https://developer.android.com/reference/android/graphics/ImageFormat#RAW_SENSOR

Capabilities and formats must be observed on the actual installation. The specification remains portable by representing absent features and unsupported profiles explicitly.

61 Version 0.3 reference build profile

Concrete interfaces and a reproducible first execution example

Version 0.3 supplies a build specification for the architecture already defined. The production contracts remain broader than the first reference evaluator. tomoss_refbuild_v1 binds a deliberately small fixture to explicit records, a task graph and exact expected outputs, so the first implementation has an inspectable starting point.

Layer	Deliverable in this revision
Ownership and control	Module responsibilities, in-process request/response shapes, cancellation rules and a native ABI declaration; Sections 62-65.
Concrete data and plans	A restricted record profile, complete fixture manifest, world/transition records and schedule model; Sections 66-69.
Reference execution	A complete C++20 serial evaluator and measured validation results; Sections 70-72.
Evolution and deployment	Activation/migration rules, app workflow, resource admission and build acceptance artifacts; Sections 73-76.

One small implementation with a precise claim

The evaluator checks the two-entity integrity/Q16 example. It enumerates task permutations and accepts only those satisfying the declared dependency graph. The fixture uses exact integer arithmetic and exact Q16 endpoints. It does not load a model, fetch telemetry, mutate a live world, commit a database transaction or submit GPU work. Those capabilities retain their own acceptance gates.

Compatibility and naming

The reference profile refines the proposed descriptor/execution v2 contracts. Its own named records identify their restricted scope; they do not redefine every v2 implementation or change retained log net D payloads. Schema, profile, implementation and document versions are separate identities. A future broader binding receives its own version and conformance evidence.

The first complete runtime build must connect validated records to admission, execution, durable commitment, coherent publication and recoverable delivery. Passing the serial evaluator supplies evidence about its arithmetic and schedule logic only. Section 72 states exactly what was run and Section 76 names the remaining build outputs. [V3]

62 Module ownership and dependencies

Each mutable authority has one owner and a narrow interface

PROPOSED. The initial runtime exposes in-process services. Language bindings translate these contracts without adding a network protocol. Caller capabilities are checked by the service; knowing an object ID or possessing a content digest grants no authority.

Owner	Allowed writes and dependencies
Record service	Validate canonical records and retain immutable artifacts. Publish digest references only after bytes are durable; never rewrite retained payload meaning.
Registry and plan compiler	Stage modules; validate schemas, ABI, capabilities and dependency graphs; register immutable versions. Read artifacts and capability reports; never activate an instance directly.
Instance controller	Own mutation admission, state/world candidates and coherent publication for one instance. Use the registry, scheduler and writer coordinator; never expose a partial candidate.
Q0 and workers	Schedule durable READY attempts and prepare bounded private output. Read frozen plans; request status changes through the coordinator. Workers cannot write the instance head.
Writer coordinator	Serialize database writes. Atomically record admissions, terminal outcomes, state activation, journal and outbox; check expected generations for mutations.
Delivery service	Read eligible committed events; append delivery attempts and acknowledgements through the coordinator. It cannot recompute or overwrite results.
Views and host adapters	Read retained snapshots and results under explicit capabilities. Submit new intents for changes; device effects originate from committed delivery records.

Dependency direction and privilege

Numerical kernels depend on typed input and execution context, not on the controller, database or UI. The controller calls the writer; the writer does not invoke workers or callbacks inside a transaction. Delivery dispatch runs outside publisher locks. [X1-X3; Y3]

Separate grants cover read, compute, instance mutation, module staging and plan activation. Staging verified bytes does not authorize executing them. A UI binding cannot widen its caller's grants. Shared caches contain immutable derived values; record retention and authoritative recovery remain separate responsibilities.

63 Control API request and response contracts

Versioned method shapes expose identities and durable status

PROPOSED. Every call returns Result<T> or Error, using typed IDs and versioned records. CallContext carries caller capability, correlation ID and optional wait limit. A wait limit bounds this call; it does not cancel accepted work. No REST transport is assumed.

Method shape	Request and returned contract
CreateInstance(req)	Intent ID, initial descriptor and input references; return instance ID, initial snapshot/generation and durable disposition. Requires instance-create authority.
SubmitIntent(req)	Intent ID, instance, operation, input references, binding policy, budget and deadline policy; return intent status and attempt ID only when admitted.
GetAttempt(id)	Return frozen manifest/snapshot, phase, typed cause and committed result reference when present; delivery status is separate.
CancelAttempt(id, request_id)	Return cancellation disposition and observed phase. Recording a cancellation request is distinct from a durable terminal outcome.
InspectPlan(digest)	Return the immutable component/system graph, versions, bindings, schemas and validation report for that plan.
ReadSnapshot(ref)	Return an owned immutable lease to the exact instance/generation/digest; never substitute the newest snapshot.
GetEvents(cursor, limit)	Return ordered committed journal events and continuation cursor. Include stable IDs; an expired retention cursor returns cursor_expired.
StageModule(manifest)	Return staged artifact identity and validation report. Requires staging authority; no instance publication follows.
ActivatePlan(req)	Intent ID, instance, expected generation, plan and migration references; return durable status/activation identity. Requires activation authority.

Errors are typed records

Error carries code, retry_class, correlation/intent/attempt IDs when known, stage, offending reference and bounded explanation. invalid_request, unsupported and permission_denied require correction; busy and storage_unavailable permit bounded retry; state_conflict requires explicit rebinding; too_late preserves the committed outcome. not_found and cursor_expired never invent replacements.

A read reports its record version. GetEvents exposes durable history, not proof of pointer publication. Admission distinguishes pending intent from durable READY. Activation distinguishes committed generation from session publication; neither implies observer acknowledgement.

64 Intents, attempts, cancellation and deadlines

Acceptance binds one execution without merging independent requests

PROPOSED. RECEIVED records a request intent, its digest and declared binding policy. A pending mutation does not freeze an obsolete instance state while waiting for its predecessor. When eligible, the owner resolves one complete snapshot and reserves capacity before durable admission.

```
Mutation READY precondition, under the instance owner:
  pinned_base == session_published_generation == durable_head
  no unresolved mutation; capacity reserved; inputs validated
Commit READY + admission sequence + reservation + journal.
Acknowledge only after commit; Q0 notification is idempotent.
```

The same intent ID and request digest return existing status, not a fresh attempt or generation. A changed digest under that ID returns idempotency_conflict. Execution retries are linked attempts retaining the frozen dependencies. State rebinding requires an explicitly linked new intent and complete validation; duplicate submission never authorizes it.

Sharing computation does not merge ownership

A pure computation key may reuse immutable output across intents. Each intent retains its own provenance, attempt status and cancellation ownership. Cancelling one subscriber cannot cancel work still required by another. Pure attempts may commit results from retained older snapshots. Mutations cannot use content equality to merge distinct intended actions.

Three different time and lifecycle decisions

A compute deadline belongs to the execution policy. Expiry requests termination with cause deadline_exceeded; the coordinator serializes it against commit. A cancellation request is likewise a request until its winning terminal disposition is durable. Committed work answers too_late even while delivery remains pending. [Y3]

Deadline records name their clock domain and restart policy; never reuse a prior process's monotonic timestamp after reboot. A paused/backgrounded host is not automatic cancellation. Resume or terminate according to recorded policy. Accepted failed/cancelled attempts receive terminal sequence records; storage failure pauses recovery rather than claiming an unwritten terminal result.

A caller timeout or lost response requires status lookup. It does not prove rejection, rollback, cancellation or completion of device work.

65 Native ABI declaration and completion

A compact proposed C boundary carries bytes and explicit ownership

PROPOSED. Registry factories supply ta_op handles for admitted implementations. Match target ABI, calling convention and v1 layout; validate abi and bytes before callbacks. These process structs are not serialized records. Declaration syntax passed C11 and C++20 checks; implementations remain untested. [Y6; V3]

```
#include <stddef.h>
#include <stdint.h>
#ifdef __cplusplus
extern "C" {
#endif
typedef struct ta_op ta_op;
typedef struct ta_done ta_done;
typedef int32_t ta_status;
enum { TA_OK=0, TA_PENDING=1, TA_CANCELLED=2,
       TA_FAILED=-1, TA_INVALID=-2, TA_NOMEM=-3 };
typedef struct { const uint8_t *data; size_t size; } ta_bytes;
typedef struct {
  uint32_t abi, bytes; void *ctx;
  void *(*alloc)(void*, size_t size, size_t alignment);
  void (*free)(void*, void*, size_t size, size_t alignment);
  int (*cancelled)(void*);
} ta_host_v1;
ta_status ta_submit_v1(ta_op*, const ta_host_v1*,
                       ta_bytes input, ta_done** out);
ta_status ta_poll_v1(ta_done*, ta_status* operation,
                     ta_bytes* output, ta_bytes* diagnostic);
ta_status ta_done_release_v1(ta_done*);
#ifdef __cplusplus
}
#endif
```

submit returns TA_OK with a completion handle after accepting work, or a negative status with a null handle. poll is nonblocking: TA_PENDING means unfinished; TA_OK exposes terminal operation status and borrowed output/diagnostic bytes. Unsuccessful operations expose empty output and a typed diagnostic. Submission is not computation success.

The host retains input, callbacks/context, snapshot and module through completion and release. Output/diagnostic buffers use its allocator and expire at release. Releasing unfinished work returns TA_PENDING. Serialize submit per operator and poll/release per completion; allocator/cancellation callbacks must be thread-safe. Null span data is allowed only at zero length.

Completion requires all owned CPU/device work and required host visibility to finish, including cancelled work. GPU fences and module retention still apply. The host validates pointer/length/schema/alignment and handle ownership. Operators catch internal exceptions; no exception, STL object or C++ ownership type crosses this boundary. Cancellation polling never commits state. [Y5-Y6; Y11]

66 Exact record and validation contracts

A small binding profile makes the reference fixture reproducible

PROPOSED. tomoss_refbuild_v1 refines the v2 descriptor/execution contracts for the ecs_integrity_demo_v1 fixture. Its companion manifest, world and expected-transition schemas are new identities. They do not redefine v1 records, general v2 formats or retained coordinate bytes. TESTED applies only to the local checks reported in Section 72.

Profile item	Exact restriction
Identity and revision	Instance fixture-1; world fixture-world-1; entity_serial strings "41" and "42" in numeric order; next serial "43". Generations/revisions are integers 0 or 1.
IntegrityUnits v1	Integer X from 0 through 20; unit 1/20; next X = max(0, X - 1). Compare integers: X < 3 is ONSET; X = 3 is BOUNDARY; otherwise STABLE.
Q16Signal v1 binding	Retained unsigned 16-bit code domain. This evaluator supports endpoints 0 and 65535 only, with positive s0 exactly 1 in fixture_signal_unit_v1. Intermediate codes return endpoint_only.
Root state and classes	ecs_integrity_root_v1 binds instance, generation and one mode. Classification ranks STABLE, BOUNDARY, ONSET as 0, 1, 2; aggregate their maximum.
Record validation	Reject missing/unknown fields, duplicate keys, incorrect types, noncanonical serial strings and incompatible references. Booleans never substitute for integer fields.

Bounded canonicalization subset

Input is at most 65,536 ASCII bytes; nesting depth is at most 12; objects/arrays contain at most 32 members/items; strings contain at most 128 printable ASCII characters. Integers satisfy |n| <= 9,007,199,254,740,991. Reject fractional/exponent tokens, negative zero, nonfinite values and unsupported strings before coercion. Sort keys and emit compact JSON without a trailing newline.

This is a restricted JCS-compatible subset, not a general RFC 8785 implementation. RecordId uses Section 48 framing with kind = record.schema. Raw file SHA-256 has a separate meaning. Wider retained words remain strings/bytes. Full Unicode/number conformance remains an implementation gate. [Y2]

67 A concrete descriptor manifest

Complete literal bindings for one fixed reference graph

Each state_rows entry selects the common graph below under its named mode. This fixture permits BOUNDARY to ONSET. Registry namespaces are separate; version 1 definitions are fixed by Sections 66 and 69 and the complete source. The manifest is a companion binding record, not a general v2 parser implementation.

```
{
 "schema":"tomoss_refbuild_manifest_v1","binding_profile":"tomoss_refbuild_v1",
 "descriptor_contract":"tomoss_state_descriptor_v2",
 "execution_contract":"tomoss_execution_v2",
 "model_profile":"ecs_integrity_demo_v1",
 "state_rows":{"STABLE":"ref.stable.v1","BOUNDARY":"ref.boundary.v1",
               "ONSET":"ref.onset.v1"},
 "state_schema":"ecs_integrity_root_v1",
 "world_schema":"tomoss_refbuild_world_v1",
 "backend":"serial_cpp20_reference_v1","numeric_policy":"exact_integer_endpoint_v1",
 "registries":{"runtime.module":{"ref_fixture":"1"},
  "ecs.component":{"IntegrityUnits":"1","Q16Signal":"1","Classification":"1"},
  "ecs.system":{"StepIntegrity":"1","DecodeSignal":"1","Classify":"1",
                "AggregateMode":"1","MergeValidate":"1"}},
 "source_sha256":"f6bf10e56f21e36975a195c5644018045f238daeca3ba0aec979e0878908bb64",
 "world_snapshot_ref":
 "90259dece6d1ca0decbb4eec837c60cd73c543bfbd58a5bf2cd60ee909a29b88",
 "systems":["StepIntegrity","DecodeSignal","Classify",
            "AggregateMode","MergeValidate"],
 "edges":[["StepIntegrity","Classify"],["Classify","AggregateMode"],
          ["AggregateMode","MergeValidate"],["DecodeSignal","MergeValidate"]],
 "parameters":{"integrity_unit":[1,20],"integrity_range":[0,20],
  "step":1,"threshold":3,"s0":1,"signal_unit":"fixture_signal_unit_v1",
  "supported_codes":[0,65535]},
 "schedule":{"entity_order":"numeric_serial","logical_partitions":1,
  "reduction":"maximum_class_rank","class_rank":["STABLE","BOUNDARY","ONSET"],
  "merge_policy":"reject_overlap"},
 "limits":{"entities":2,"tasks":5,"classification_updates":2},
 "transitions":[["BOUNDARY","ONSET"]],"effects":[]
}
```

world_snapshot_ref is the domain-separated RecordId of the complete initial world on the next page. source_sha256 identifies the raw C++ source bytes. Display whitespace does not belong to canonical record identity.

Manifest RecordId

```
5bf4aa40a6a91cd7816e582641e84a04f41151df5f3fb044286f9f5be5d0c0ec
```

68 World snapshot and transition records

Complete fixture literals preserve identity across the expected step

ILLUSTRATIVE records, validated locally. Only integrity, classification, revision and generation change. The expected-transition record states explicitly that durable commit and publication were not executed. Its world/manifest references are Section 48 RecordIds.

Initial world


```
{"schema":"tomoss_refbuild_world_v1","binding_profile":"tomoss_refbuild_v1",
 "instance_id":"fixture-1","world_id":"fixture-world-1","generation":0,
 "next_entity_serial":"43","entities":[
 {"entity_serial":"41","revision":0,"integrity_units":4,
  "q16_code":0,"class":"STABLE"},
 {"entity_serial":"42","revision":0,"integrity_units":3,
  "q16_code":65535,"class":"BOUNDARY"}]}
```


Expected next world


```
{"schema":"tomoss_refbuild_world_v1","binding_profile":"tomoss_refbuild_v1",
 "instance_id":"fixture-1","world_id":"fixture-world-1","generation":1,
 "next_entity_serial":"43","entities":[
 {"entity_serial":"41","revision":1,"integrity_units":3,
  "q16_code":0,"class":"BOUNDARY"},
 {"entity_serial":"42","revision":1,"integrity_units":2,
  "q16_code":65535,"class":"ONSET"}]}
```


Expected transition


```
{"schema":"tomoss_refbuild_expected_transition_v1",
 "binding_profile":"tomoss_refbuild_v1","evidence":"expected_fixture_record",
 "instance_id":"fixture-1","logical_transition_id":"fixture-step-1",
 "base_generation":0,"next_generation":1,
 "base_world_sha256":
 "90259dece6d1ca0decbb4eec837c60cd73c543bfbd58a5bf2cd60ee909a29b88",
 "next_world_sha256":
 "bce0f9308ab4dabeed719141633d46e6b2ee1d2f6c462265cd34f24b85ab4fa9",
 "manifest_sha256":
 "5bf4aa40a6a91cd7816e582641e84a04f41151df5f3fb044286f9f5be5d0c0ec",
 "root_state_before":"BOUNDARY","root_state_after":"ONSET",
 "signals":[{"entity_serial":"41","value":0,"unit":"fixture_signal_unit_v1"},
            {"entity_serial":"42","value":15,"unit":"fixture_signal_unit_v1"}],
 "outcome":"VALID_CANDIDATE","durable_commit":"not_executed",
 "publication":"not_executed"}
```


The C++ program hardcodes the two numerical entities and computes their candidate outputs. Python separately validates these companion JSON records and identities. The C++ program does not parse JSON, persist snapshots or instantiate an ECS library.

69 Scheduler reference model

Exhaustive serial order evaluation tests a bounded dependency graph

PROPOSED reference semantics. A task may run once, after all predecessors. StepIntegrity precedes Classify, Classify precedes AggregateMode, and both AggregateMode and DecodeSignal precede MergeValidate. Enumerate every permutation of the five distinct tasks: 120 orders, exactly four satisfying these dependencies.

Task	Reads	Private output or validation
StepIntegrity	Base integrity units	Next integrity; subtract one with zero clamp.
DecodeSignal	Base Q16 codes; s0 = 1	Exact signal endpoints 0 and 15.
Classify	Next integrity	Per-entity categorical classes.
AggregateMode	New classes	Maximum declared class rank.
MergeValidate	Candidate world; predecessor completion	Verify candidate bounds after all predecessors.

The four accepted orders

```
StepIntegrity -> DecodeSignal -> Classify -> AggregateMode -> MergeValidate
StepIntegrity -> Classify -> DecodeSignal -> AggregateMode -> MergeValidate
StepIntegrity -> Classify -> AggregateMode -> DecodeSignal -> MergeValidate
DecodeSignal -> StepIntegrity -> Classify -> AggregateMode -> MergeValidate
```

All four must produce integrity units [3,2], signals [0,15], classes [BOUNDARY,ONSET] and root ONSET. The base remains [4,3]. The program checks the full Result and the unchanged base after every permutation; dependency violations must report dependency, not become successful candidate results.

What this executable establishes

TESTED: the serial evaluator accepted four orders, rejected 116, and passed 12 further rejection cases. Those cases cover integrity/code bounds, nonpositive or unsupported scale, an intermediate code, invalid/duplicate identity, duplicate task and invalid task ID.

This is a serial schedule oracle with fixed-size value copies. MergeValidate models the terminal dependency/validation point; it does not implement command-buffer conflict resolution, transactional publication, real threads, general queries or device lifetimes. Those remain the separate acceptance gates in Sections 39-43 and 48-51.

70 C++ reference implementation

Part 1 of 2: types input validation and candidate evaluation

TESTED source, reproduced completely across these two pages. Concatenate the two C++ listings into reference_fixture.cpp, preserving line breaks and one final newline. The hash and build commands are separate. The fixture uses bounded integers and private value copies, without floating-point arithmetic.

```
#include <algorithm>
#include <array>
#include <cstdint>
#include <iostream>
#include <stdexcept>
#include <string>
using I = std::int64_t;
enum Task { Step, Decode, Classify, Aggregate, Merge };
enum Mode { STABLE, BOUNDARY, ONSET };
struct Entity { I serial, integrity, code;
  bool operator==(const Entity&) const = default; };
using World = std::array<Entity, 2>; using Order = std::array<int, 5>;
struct Result {
  World world; std::array<I, 2> signal{}; std::array<int, 2> classes{}; int mode{};
  bool operator==(const Result&) const = default;
};
struct Invalid : std::runtime_error { using std::runtime_error::runtime_error; };
void need(bool ok, const char* why) { if (!ok) throw Invalid(why); }
void check(bool ok) { if (!ok) throw std::logic_error("check_failed"); }
I endpoint(I code, I scale) {
  need(code >= 0 && code <= 65535, "code_range");
  need(scale > 0, "scale_nonpositive"); need(scale == 1, "unsupported_scale");
  need(code == 0 || code == 65535, "endpoint_only"); return code == 0 ? 0 : 15;
}
Result run(const World& base, const Order& order, I scale = 1) {
  for (auto e : base) {
    need(e.serial > 0, "identity");
    need(e.integrity >= 0 && e.integrity <= 20, "integrity_range");
    (void)endpoint(e.code, scale);
  }
  need(base[0].serial < base[1].serial, "identity_order");
  constexpr int prereq[5] = {0, 0, 1 << Step, 1 << Classify,
                             (1 << Aggregate) | (1 << Decode)};
  int done = 0; Result out{base};
  for (int task : order) {
    need(task >= Step && task <= Merge, "task_id");
    need(!(done & (1 << task)), "duplicate_task");
    need((done & prereq[task]) == prereq[task], "dependency");
    if (task == Step) for (auto& e : out.world)
      e.integrity = std::max<I>(0, e.integrity - 1);
    if (task == Decode) for (int i = 0; i < 2; ++i)
      out.signal[i] = endpoint(base[i].code, scale);
```

Raw source SHA-256

```
f6bf10e56f21e36975a195c5644018045f238daeca3ba0aec979e0878908bb64
```

71 Reference implementation continuation

Part 2 of 2: complete evaluation and rejection checks

```
    if (task == Classify) for (int i = 0; i < 2; ++i) {
      I x = out.world[i].integrity;
      out.classes[i] = x < 3 ? ONSET : (x == 3 ? BOUNDARY : STABLE);
    }
    if (task == Aggregate) out.mode = std::max(out.classes[0], out.classes[1]);
    if (task == Merge) for (auto e : out.world)
      need(e.integrity >= 0 && e.integrity <= 20, "candidate_range");
    done |= 1 << task;
  }
  need(done == 31, "incomplete_schedule"); return out;
}
template<class F> void rejects(F fn, const char* why) {
  try { fn(); } catch (const Invalid& e) { check(std::string(e.what())==why); return; }
  throw std::logic_error("expected_rejection");
}
int main() {
  const World base{{{41, 4, 0}, {42, 3, 65535}}}; const World saved = base;
  const Result expected{World{{{41, 3, 0}, {42, 2, 65535}}},
                        {0, 15}, {BOUNDARY, ONSET}, ONSET};
  Order order{Step, Decode, Classify, Aggregate, Merge}; int legal = 0, bad = 0;
  do {
    try { check(run(base, order) == expected); ++legal; }
    catch (const Invalid& e) { check(std::string(e.what()) == "dependency"); ++bad; }
    check(base == saved);
  } while (std::next_permutation(order.begin(), order.end()));
  check(legal == 4 && bad == 116);
  auto field = [&](int kind, I value) { World w = base;
    (kind == 0 ? w[0].integrity : w[0].code) = value; return run(w, order); };
  for (I x : {-1, 21}) rejects([&] { field(0, x); }, "integrity_range");
  for (I q : {-1, 65536}) rejects([&] { field(1, q); }, "code_range");
  for (I s : {0, -1}) rejects([&] { run(base, order, s); }, "scale_nonpositive");
  rejects([&] { run(base, order, 2); }, "unsupported_scale");
  rejects([&] { field(1, 1); }, "endpoint_only");
  World w = base; w[0].serial = 0; rejects([&] { run(w, order); }, "identity");
  w = base; w[1].serial = 41; rejects([&] { run(w, order); }, "identity_order");
  rejects([&] { run(base, Order{0, 1, 2, 3, 3}); }, "duplicate_task");
  rejects([&] { run(base, Order{0, 1, 2, 3, 5}); }, "task_id");
  check(base == saved);
  std::cout << "{\"legal_orders\":4,\"dependency_rejections\":116,"
               "\"edge_rejections\":12,\"old_world_unchanged\":true}\n";
}
```

Build and run

```
g++ -std=c++20 -O2 -Wall -Wextra -Wpedantic -Werror \
  reference_fixture.cpp -o reference_fixture
./reference_fixture
```

The exhausted permutation loop restores ascending task order before the rejection checks. Any unexpected result or rejection cause terminates unsuccessfully. The captured compiler, host, exit status and exact test counts appear in Section 72. No actual thread scheduling, publication, SQLite, Android or JIT execution is claimed.

72 Executed reference evidence

Measured results for the serial fixture and restricted record validator

VERIFIED [V3]. The complete C++ source in Sections 70-71 was compiled and executed on the authoring host. A separate Python validator checked the accompanying restricted JSON records. The executable hardcodes the fixture; it does not parse JSON or instantiate an ECS library.

Check	Observed outcome
C++ build and execution	Compile exit 0; run exit 0. Compiler diagnostics empty under warnings-as-errors.
Task permutations	120 examined: 4 dependency-legal orders; 116 rejected for unmet dependencies.
Exact output and isolation	Each legal order produced integrity [3,2], signals [0,15], classes [BOUNDARY,ONSET] and root ONSET. Original world values stayed unchanged.
Targeted C++ rejection cases	12 passed: invalid ranges, scales, endpoints, entity identity and task identifiers/duplicates.
Restricted JSON checks	4 valid records and canonical round trips; 30 targeted invalid cases rejected, including duplicate keys, boolean substitution and overlong numeric tokens.

Build environment: g++ (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0; Linux x86_64; Python 3.12.14. This is host evidence, with no Android build or latency/energy measurement.

Reproduce the C++ check

```
g++ -std=c++20 -O2 -Wall -Wextra -Wpedantic -Werror \
  reference_fixture.cpp -o reference_fixture
./reference_fixture
```

Raw SHA-256 source identities, distinct from the domain-separated record identities in Section 48:

```
reference_fixture.cpp
f6bf10e56f21e36975a195c5644018045f238daeca3ba0aec979e0878908bb64
run_reference_validation.py
966d0f205f249d5fe3ad22e01237b9227fd0129b326426f55e02c4ca8de7d240
```

The record checker covers the documented printable-ASCII/safe-integer subset only. General JCS, arbitrary Q16 decoding, actual threads, publication, SQLite, Android, JIT, GPU and calibrated physical models were not tested by this evaluator. Their acceptance requirements remain in Sections 30, 50-57 and 73-76.

73 Activation, migration and recovery contract

A configuration change is an owned durable state transition

PROPOSED. ActivatePlan is a control intent using the same instance owner and mutation slot as model transitions. Module staging only prepares immutable dependencies. Migration is a named operation with input/output schemas, implementation digest and explicit identity/quantity preservation rules.

Boundary	Required decision and evidence
Bind	Wait until eligible; validate the requested expected generation against pinned, session-published and durable heads. Freeze old snapshot, target plan and migration manifest.
Prepare privately	Construct next root state and ECS world; retain existing EntityRefs where semantics require continuity. No published object is edited and no external action occurs.
Validate	Check schemas, exact-value rules, legal transitions, entity/query invariants, bindings and capabilities. Reserve coexistence of old/new worlds, modules and migration scratch.
Commit	Serialize cancellation/status checks; require exactly one expected-generation head update. Atomically record candidate state/world/snapshot, result, terminal disposition, activation, provenance and required outbox.
Publish	Keep mutation admission closed until the coherent pointer and session publication generation advance. Release eligible notifications afterward; their immutable generation remains explicit.
Recover and retire	Reconstruct the committed head before reopening its gate. Retain older resources through all readers and device submissions; then apply retention policy.

Failure preserves the authoritative decision

Candidate validation failure leaves the previous head usable. A crash after activation commit requires reconstruction of that committed generation, not rerunning migration against newer inputs. Missing required artifacts fault that instance with a named dependency; recovery cannot silently choose another module or roll back durable history. [Y3-Y5]

Rollback is a new activation

Selecting an older compatible configuration creates a new increasing generation and a recorded state mapping. It does not erase prior results, reuse their generation or undo device actions. Acceptance must cover incompatible migration, stale expected generation, exhausted coexistence budget, crashes around publication and recovery with missing artifacts. No runtime conformance is claimed here.

74 Application modules and operator workflow

Proposed application views make source status and results inspectable

The Android controller shall coordinate a native core, durable data journal, source adapters and renderer. The operator selects sources, reviews admissibility, chooses a world and runs a declared plan. Screens present availability, progress and committed outcomes; detailed versions and digests belong in inspection views.

Module boundary	Owned responsibility
android/app + native bridge	Lifecycle, platform permissions, operator actions and bounded native calls; keep work off the UI thread.
core + runtime	Types, numerical profiles, worlds, plan construction, admission and private execution.
journal	Accepted attempts, authoritative results, snapshots, activation and delivery intents.
adapters	Explicit source access, acquisition envelopes, import staging and disconnected capture.
render	Committed world/result views with generation, quality and backend provenance.

Operator view	Action and meaningful feedback
Inspect Source / Import Review	Select a source; inspect origin, type, completeness and required access; show admitted, pending or rejected with a cause.
World / Graph Inspection	Select an instance; inspect its current state and actual active graph; compare a proposed change before activation.
Run Results / Errors	Start, observe or request cancellation; inspect committed results, input provenance and a specific failure/retry outcome.
Capabilities / Budgets	Show available sources/backends, capacity in use, queued work and why another run cannot start.

Capture and import remain explicit

Capture runs only through a permitted host lifecycle, including foreground operation when required. Disconnection uses bounded local buffers; retain source clock/session, acquisition time, receipt time and gap metadata. Reconnection preserves event identity. If capture must stop or overflow, record the gap under the selected policy. [X12-X16; Y6; Y9; Y12]

App integration requires a supported authorized interface; installing TOMOSS.A.D. does not expose every phone app. Import review admits a verified model/data manifest through Section 46. A repository selection never executes its contents. [Y6-Y7; proposed workflow]

75 Resource admission and deployment profiles

Proposed limits belong to named owners and aggregate reservations

A BudgetProfile shall bind an ID/version, capability report and explicit limits. Each reservation records owner, quantity, unit and lifetime. The admission coordinator checks aggregate committed reservations plus the candidate peak before durable acceptance. Unknown required capabilities return unavailable; the marketed 12 GB never substitutes for an observed process allowance.

Limit field / unit	Account owner	Reservation and release
resident_bytes / bytes	Coordinator	Aggregate unique allocations and safety margin; reject when the candidate peak exceeds capacity.
world_bytes; retained_generations / bytes; count	Instance owner	Reserve candidate and retained worlds; release old chunks only after their last reader/device user.
queue_jobs; queue_bytes / count; bytes	Q0	Reserve on admission; transfer accounting into execution or release after durable terminal disposition.
model_bytes; kv_bytes; workspace_bytes / bytes	Inference host	Reserve weights, KV and peak workspace before load/run; release when their owners finish.
jit_peak_bytes; table_bytes / bytes	Compiler / registry	Include C++ frontend, IR, code and staging peaks plus immutable lookup tables; retain admitted modules until unused.
gpu_bytes; gpu_submissions / bytes; count	GPU backend	Reserve buffers and in-flight work; release after completion and required visibility operations.
capture_bytes; journal_bytes / bytes	Adapter / store	Separate RAM buffers from durable storage; release reservations by consumption or eligible retention cleanup.

Count shared allocations once. Per-job limits do not bound concurrent instances. Admission also fixes context_tokens, output_reserve_tokens and retrieval_bytes; Section 47 supplies finite context and KV estimates. A failed reservation returns resource_limit without changing the active world. Cancellation does not prematurely free device or snapshot resources. [Y11; Y13; proposed accounting]

Profile activation

host_reference selects portable CPU execution; android_p0 selects packaged kernels after app/ABI gates. android_gpu, android_ai and android_jit_p2 each require their own capability evidence, resource measurements and conformance. P1 interpretation remains separately bounded. A missing optional profile stays unavailable; fallback must preserve the requested numerical contract and record its backend. [X13-X16; Y6-Y7; Y11]

76 Build backlog and acceptance package

Proposed artifacts advance only after their dependency gates pass

The names below select future source/build boundaries; they do not describe an existing Android build. Preserve the serial reference and its evidence separately from the production runtime. Each package shall publish its source identity, dependency versions, build configuration and gate results before a dependent package claims conformance.

Package and proposed output	Dependency and exit gate
A. core/, records/ -> libtomoss_core.a	Start from retained profiles and serialized fixtures. Pass exact codec boundaries, malformed-input rejection and canonical identities.
B. runtime/, conformance/ -> tomoss_runtime_cli	Depends on A. Execute typed plans, immutable worlds and cancellation/migration; test old readers, conflicting writers and multiworker replay.
C. journal/ -> libtomoss_journal.a	Depends on A-B. Pass admission/commit crash injection, recovery, cancellation races and delivery reconciliation against the selected SQLite build.
D. android/ -> libtomoss_jni.so; tomoss-debug.apk	Depends on A-C. Pass native bridge, lifecycle/process-death recovery, committed-result views and measured device admission.
E. adapters/, render/, compiler/ -> profile bundles	Depends on the relevant A-D gates. Admit pinned imports, capture, GPU and P1/P2 separately with their own fixtures and budgets.

Established evidence and remaining gates

Sections 70-72 record the small serial C++ reference: four dependency-legal task orders among 120 permutations, producing integrity 3 and 2 and signals 0 and 15. This verifies the selected serial fixture. Threaded scheduling, SQLite transactions, Android deployment and native JIT remain pending implementation gates; those outcomes cannot be inferred from serial order enumeration.

Acceptance package

Collect source/artifact digests, reproducible build commands, schemas, descriptor/input fixtures, expected and observed outputs, exit statuses and raw reports. Add race/crash traces, unsupported-capability results, peak resource measurements and recovery instructions as those implementations exist. Record exact versus tolerance profiles and the tested platform.

A release manifest links every capability to its passing evidence and labels absent gates pending. An application artifact enters the package only after it has actually been built and tested. No performance, universal phone access or physical-model validity follows from a successful compilation. [Sections 30, 49-57, 62-73]

77 Reference build sources and evidence

How to reproduce the new checks and interpret their scope

[Z1] JSON grammar and interoperability

Bray, RFC 8259, The JavaScript Object Notation (JSON) Data Interchange Format. JSON grammar and interoperability considerations, including object names and numerical representation. https://www.rfc-editor.org/rfc/rfc8259

The fixture validator imposes a narrower documented profile. Syntactic JSON validity alone does not establish schema validity, component semantics, numerical exactness or capability authority. General canonicalization remains governed by RFC 8785 [Y2], not by assuming that every generic JSON serializer implements it.

[Z2] C++ permutation enumeration

ISO C++ working draft, Permutation generators. The next_permutation operation advances a range in lexicographic permutation order and reports exhaustion. https://eel.is/c++draft/alg.permutation.generators

The reference starts from the ordered five-task sequence. Its explicit dependency predicate and expected outputs are TOMOSS.A.D. fixture requirements. The compiler invocation and source identity are recorded with the executed evidence, rather than inferred from the language label.

[V3] Local reference evidence

Sections 66-72 provide the concrete profile, fixture data, complete C++ source and reported checks produced during this revision. This evidence is distinct from the retained numerical bundle verification [V], the attachment algebra/code checks [V2], and the proposed runtime acceptance suite.

Reproduction must use the complete source across both source pages, the declared compiler mode and the unchanged fixture contract. A modified source, manifest, numerical policy or compiler profile yields a new evidence record. Source digests identify bytes; they are not an assurance that a different build will have identical behaviour.

Existing primary references remain applicable

ECS and immutable ownership: [X1-X4; Y1]. Canonical identity and durability: [Y2-Y4]. Compiler and model admission: [Y5; Y7-Y8; Y13]. Android lifecycle and native boundaries: [X12-X16; Y6]. Telemetry, terrain and rendering: [Y9-Y12]. The reference extension does not claim new device access or an admitted external artifact.

The build backlog closes the remaining gap with targeted race, crash, module-lifetime, device and workload evidence. Every capability claim should point to a named implementation and the acceptance result that established it.